1. Starting points

Axioms

1. **Axiom.** Attending a WG21 meeting in person costs money and time.
2. **Axiom.** A sponsor can pay a participant's travel and salary, but cannot attend, speak, or vote in the participant's place.
3. **Axiom.** A participant who attends most meetings expects to meet many of the same colleagues at future meetings. A chair expects fewer problems from advancing a paper whose author has a record of following the procedures and responding to review.
4. **Axiom.** Given a choice, and other things equal, each participant, sponsor, national body, and user prefers the option that brings it more of what it values, sooner, at less cost to it, and weighs a gain in one value against a loss in another on one scale of its own. For this proof, a participant values adoption of the proposals they support, the interests of their sponsor if they have one, their standing, keeping any chair or convenor post they hold, their time and money, and a good result for users; a sponsor values its interests, which can depend on particular features of a design; a national body values the outcomes its national position supports and, where it has none yet, the interests concerned at its national level; and a user values the features they use or would use, how well those features work, and their own time and money.
5. **Axiom.** A participant who votes Against a colleague's proposal cannot rule out that the colleague will support the participant's own proposals less readily. A Neutral vote, or not voting, gives the colleague no more reason for this than an Against vote does.
6. **Axiom.** No participant can study every paper closely, so each chooses which papers to study, and what a participant learns from a paper or its discussion can change their vote.
7. **Axiom.** Writing or reviewing a paper takes time, and, other things equal, a longer or more complex paper takes more; preparing to give reasons for a vote also takes time.
8. **Axiom.** Until a design is implemented and used, nobody can be sure how well it works, and even then those voting on it cannot fully judge how well it serves users. A design with more parts and more interactions with existing wording has more places where a defect can go unnoticed.
9. **Axiom.** When a proposal is first presented, it has received less committee review than at any later meeting. When a design is presented in parts, reviewers of an early part cannot study in detail what the later parts will require of it until those parts are presented.
10. **Axiom.** If a decision is reversed, the work built on it must be redone or discarded. After a favorable poll, authors and others build more on the design, such as wording, implementations, and proposals that depend on it.
11. **Axiom.** Revising a design throughout costs its authors more work than a change confined to one part of it.
12. **Axiom.** A participant learns how the committee works partly by attending, some of it only that way, and this knowledge differs from skill at design. Participants can use only procedural options they know,

and, other things equal, those who know more spend less effort getting a paper considered and have a better chance of adoption.

13. **Axiom.** Over the years, some participants stop attending and others join, while a published paper stays available to later readers for decades. Someone absent from a meeting learns what was said there only from writing or from those present who remember it, and a clear written objection is easier to raise again.
14. **Axiom.** A meeting has a fixed number of hours, so time given to one paper is not available to another.
15. **Axiom.** WG21 plans a new edition of the C++ standard every three years, and ISO publishes each edition once national bodies approve it.
16. **Axiom.** When several subgroups meet at the same time, a participant can attend only one of them.
17. **Axiom.** A design in the standard applies to every user of its feature, so a better design's gain is spread across them all, while participating costs the participant or their sponsor. Users differ from each other and from participants in the features they care about and what they can spend taking part, and one user's effort seldom changes a committee outcome.

Definitions

18. **Definition.** A participant is a registered expert attending a WG21 meeting. Guests a convenor invites are not participants.
19. **Definition.** A regular is a participant who expects to attend future meetings with many of the same colleagues.
20. **Definition.** A newcomer is a participant attending for the first time; a newcomer who expects to attend future meetings with many of the same colleagues is also a regular.
21. **Definition.** An author is a participant who presents a proposal paper as one of its authors.
22. **Definition.** A sponsor is an organization that pays some or all of a participant's costs of taking part.
23. **Definition.** A user is a person who writes C++ and does not participate.
24. **Definition.** A design room is EWG or LEWG, a study group is any WG21 study group, a wording room is CWG or LWG, and a subgroup is any of these. Plenary is a WG21 meeting's closing plenary session, a chair chairs a subgroup, the convenor convenes WG21, and a consensus call is a chair's or the convenor's determination of consensus.
25. **Definition.** A national body is an ISO member body that is a P-member of SC22, the committee WG21 belongs to. In quotations from the Directives, "the committee" is SC22, "the Chair" its chair, and "National Body" any ISO member body, which on enquiry and final ballots includes non-P-members of SC22.
26. **Definition.** Favor, Neutral, and Against are the poll positions as SD-4 capitalizes them, with Strongly Favor and Weakly Favor counting as Favor and likewise for Against. The 2:1 norm is SD-4's statement on subgroup polls that a proposal normally advances when, after discussion of the concerns of those voting against, more than twice as many are in favor as against, even if many vote Neutral, though a Neutral majority of those voting can be concerning.

27. **Definition.** A participant's standing is the readiness of colleagues to support the participant's future proposals.
28. **Definition.** A player's stake in a poll is the difference the poll's outcome makes to what the player values.
29. **Definition.** A statement that a player does better with one option than another means the option is worth more to the player, by their weighing of the values the statement names. Every value the statement does not name is held equal.
30. **Definition.** A direction poll is a poll on whether to continue work on a design, and a forwarding poll is a poll to send a proposal to the next room.
31. **Definition.** An incumbent on a topic is a design that has passed a direction poll in a design room and has not failed a later direction poll there.
32. **Definition.** An edition is one published version of the C++ standard, and its feature cutoff is the last meeting at which a new feature can be approved for it.
33. **Definition.** A reversal is the removal or replacement of a design or decision that WG21 or one of its subgroups has approved.
34. **Definition.** A player is a person or body whose choices the proof considers. A settled pattern, which game theory calls a Nash equilibrium, is a set of choices, one per player, in which no player gains by changing only their own choice.
35. **Definition.** A small proposal is a proposal by one author who pays their own costs and expects its design to be reviewed within one meeting.
36. **Definition.** A large proposal is a proposal whose two or more authors have their costs paid by one or more sponsors and expect its review to span more than one edition. Its paper is longer and more complex than the paper of any small proposal.
37. **Definition.** A result-seeking author is an author whose gain from a proposal depends on how well the adopted design serves its users, not on whether it keeps any particular feature.
38. **Definition.** A design-bound author is one whose gain from a proposal depends on the adopted design keeping particular features fixed before review, for example because a sponsor's interests depend on them.

Facts

39. **Fact.** SD-4's section "Purpose and audience" contains these statements.
    a. **Fact.** "This document summarizes some of WG21's current practices and procedures, in addition to the requirements of the ISO/IEC Directives and JTC 1 Supplement available at the ISO Directives and Policies page." "Note: This document does not repeat the basic material already covered in the following:" (SD-4, "Purpose and audience")

b. **Fact.** "Although this document is also useful to new participants to get up to speed, the material herein covers important procedures and situations that arise regularly in our meetings, email lists, formal ballots, and other participation, and everyone who participates in WG21 is expected to be familiar with this information." (SD-4, "Purpose and audience")

40. **Fact.** SD-4's section "Code of Conduct" contains these statements, all but the first from a slide it displays.
    a. **Fact.** "To report a concern about a potential Code of Conduct violation, follow the steps in the ISO document “Guidance and Process For Addressing Misconduct and Breaches of the Code of Conduct.”" (SD-4, "Code of Conduct")
    b. **Fact.** "Act in good faith & with due care & diligence" (SD-4, "Code of Conduct")
    c. **Fact.** "Avoid collusive or anticompetitive behavior" (SD-4, "Code of Conduct")
    d. **Fact.** "Respect others & their opinions" (SD-4, "Code of Conduct")
    e. **Fact.** "Accept group decisions" (SD-4, "Code of Conduct")
    f. **Fact.** "Ensure that all views are heard & understood" (SD-4, "Code of Conduct")
41. **Fact.** SD-4's section "Consensus" contains these statements.
    a. **Fact.** "consensus: General agreement, characterized by the absence of sustained opposition to substantial issues by any important part of the concerned interests and by a process that involves seeking to take into account the views of all parties concerned and to reconcile any conflicting arguments." "NOTE Consensus need not imply unanimity." (SD-4, "Consensus")
    b. **Fact.** "Consensus is not a tyranny of the majority. The majority cannot just say “we have enough votes” and refuse to listen to the concerns of the minority. The ISO Code of Conduct requires the majority to 'ensure that the views of all are heard and understood.'" (SD-4, "Consensus")
    c. **Fact.** "Consensus is not a tyranny of the minority. The minority, once they have had opportunity to be heard, cannot just say “we still object so you can't proceed” and refuse to accept the consensus of the majority to pursue a different direction. The ISO Code of Conduct requires the minority to 'accept group decisions.'" (SD-4, "Consensus")
    d. **Fact.** "Consensus does not preclude judicious escalation (“throwing an exception”). In rare cases, where the minority still believe that a serious error of judgment is being made, escalation can be appropriate in accord with the Escalation procedures covered later in this document; note that the key words are “rare” (not regular), “still” (it was raised before), and “serious.”" (SD-4, "Consensus")
    e. **Fact.** "Escalation can become inappropriate in two ways: (a) when a serious objection is not raised “in a timely manner,” such as being withheld until plenary session or formal ballot time rather than being raised consistently throughout the process starting as early as possible; or (b) when a participant or national body regularly uses the escalation process to express a pattern of strong disagreement on topic after topic, which erodes their credibility and is not the purpose of the escalation resolution process (like exception handling, escalation handling is for hard errors, and is not designed for expressing less serious conditions or for what should be ordinary control flow)." (SD-4, "Consensus")
42. **Fact.** SD-4's section "Proposals and papers" contains these statements.

a. **Fact.** "If a proposal doesn't have a paper, it doesn't exist. Any significant proposal requires a paper, including every proposal that is a design addition or change." "Note: A paper is not needed for minor issues that are about improving standardese wording to implement the decided design. Those do not affect the design itself, and can be put on issues lists only." (SD-4, "Proposals and papers")
b. **Fact.** SD-4's list under "A good proposal paper should:" includes "Be principle-based: Articulate the design principles for the proposed solution. Show how the proposed solution fits with the rest of the language's principles and design philosophy (note: explicitly not with the language's quirks)." (SD-4, "Proposals and papers")
c. **Fact.** SD-4's list under "A good proposal paper should:" includes "Show design alternatives considered: along with concrete examples showing why they were not pursued." (SD-4, "Proposals and papers")
d. **Fact.** "All papers to be considered at a meeting must be received before that meeting's pre-meeting mailing deadline. The agenda for each meeting includes all papers in mailings since the previous meeting (from the post-meeting mailing of the previous meeting to the pre-meeting mailing of this meeting, inclusive), as well as possibly selected earlier papers that were not sufficiently addressed at a previous meeting in the judgment of the subgroup chair." (SD-4, "Proposals and papers")
e. **Fact.** "Corollary: If a topic does not have at least one on-time paper, the whole topic is not on the meeting agenda." "Per ISO rules, if a WG meeting were to take action on a topic that was not mentioned on the agenda (including in the topics of on-time papers), a national body could formally escalate an objection in SC 22 and JTC 1 to the entire WG meeting on the grounds that the national body had insufficient notice so as to be able to send any (or the right) participants to participate in that technical discussion topic." (SD-4, "Proposals and papers")
f. **Fact.** "A proposal that does not have an on-time paper published since the previous meeting is not actionable and will not be considered during official meeting time; the author is encouraged to submit a revision as an on-time paper for a future meeting." (SD-4, "Proposals and papers")
g. **Fact.** "Followup papers to an on-time paper, such as late or in-meeting rebuttal/elaboration/update papers to an on-time paper, can of course be considered; they are part of considering the on-time paper." (SD-4, "Proposals and papers")
h. **Fact.** "A proposal that does not have one of its authors, or another qualified and prepared presenter, to present it is not actionable and will not be considered during official meeting time." (SD-4, "Proposals and papers")
i. **Fact.** "The purpose of a presentation is for a person who is an expert in the paper's domain (including in constraints as well as design alternatives considered) to educate a group of general C++ experts." (SD-4, "Proposals and papers")
j. **Fact.** "The presenter must be someone who is invested in the proposal and will ensure feedback will be used, which means the presenter will personally produce a new revision of the paper based on feedback or else will communicate feedback to the author to do so." (SD-4, "Proposals and papers")

k. **Fact.** "We cannot act on ideas without papers, and we do not significantly delay progress on concrete proposals in order to wait for alternative proposals we might get in the future. If we have an active proposal that is making progress, and an objection is raised that a different competing approach would be better but that other approach does not yet have a paper, the EWG or LEWG subgroup chair may elect to delay the progress of the active proposal by one meeting to give the objector(s) opportunity to bring an on-time proposal paper. Otherwise, if a competing alternative does not have a paper, it does not exist and will not block progress of a proposal that we do have before us." (SD-4, "Proposals and papers")

l. **Fact.** "At early discussion phases, subgroup chairs have discretion to allow limited exploration of alternatives to an active proposal to decide whether a second paper is warranted, or to discuss broader prerequisite topics that are exposed by a paper under discussion." (SD-4, "Proposals and papers")

m. **Fact.** "If a proposal has consensus to be in a TS, but does not yet have consensus to be in the IS, that means its progress is delayed by approximately 1-2 years; that is the time it takes to publish the TS, and then wait for usually one year before the TS is proposed to be merged into the IS with any changes based on the TS experience." (SD-4, "Proposals and papers")

n. **Fact.** "If by the time the TS is ready to be considered for merging into the IS we do not have alternative proposals actively progressing, the default action to is to move forward with the proposal we have, we will not wait indefinitely for something better." (SD-4, "Proposals and papers")

43. **Fact.** SD-4's section "Subgroup responsibility and polls" contains these statements.

a. **Fact.** "Subgroup chairs are appointed by the convener, and are selected to match the current needs of the subgroup. They have no fixed term." (SD-4, "Subgroup responsibility and polls")

b. **Fact.** "In subgroup polls, each person in the room can cast one vote." (SD-4, "Subgroup responsibility and polls")

c. **Fact.** "The subgroup chair may take any polls they choose." (SD-4, "Subgroup responsibility and polls")

d. **Fact.** "Multiple polls, where multiple alternatives are polled (e.g., by 5-way poll) in order to see which has most consensus. This is encouraged so as not to lose the option that has the most consensus because of the order in which polls are taken." (SD-4, "Subgroup responsibility and polls")

e. **Fact.** "Subgroup polls, especially in design subgroups, should favor progress." (SD-4, "Subgroup responsibility and polls")

f. **Fact.** "A proposal normally advances if there are more than twice as many in favor of a proposal as against, after discussion of the concerns of those voting against and possibly a re-poll to see if opinions have improved. This is true even if a large number vote Neutral, though it can be concerning if a majority of all those voting vote Neutral." (SD-4, "Subgroup responsibility and polls")

g. **Fact.** "Not everyone in the room needs to vote (even Neutral). Not everyone is interested in every proposal, and participants who are not familiar with the material of a particular poll (did not read the paper and/or were not closely paying attention to the discussion) typically do not vote on that poll." (SD-4, "Subgroup responsibility and polls")
h. **Fact.** "Anyone voting Against, especially Strongly Against, may be asked for their reasons by the chair. They should be prepared to articulate their rationale, including either what change(s) would address their concern and change their position, or else that they are opposed to pursuing the proposal in any form and no change would change their position." (SD-4, "Subgroup responsibility and polls")
i. **Fact.** "The direction group maintains a list of the proposals it considers the most important for the next version of C++ or to otherwise make progress such as in a TS, and the design group chairs use that that list to prioritize work at meetings." (SD-4, "Subgroup responsibility and polls")
j. **Fact.** Of the direction group, SD-4 says "Their charter includes setting forth a group opinion on:" and lists "Providing an opinion on any specific proposal: This includes whether the proposal should be pursued or not, and if pursued any changes that should be considered. Design group participants are strongly encouraged to give weight to an opinion that the design group feel strongly enough about to suggest." (SD-4, "Subgroup responsibility and polls")
k. **Fact.** "Study groups (SGs) own “incubating” to produce an initial proposal that can then enter the usual design group processing." (SD-4, "Subgroup responsibility and polls")
l. **Fact.** "An SG is formed by the convener at the recommendation of a design subgroup chair when there appears to be broad interest in a topic, there is not yet a unified initial proposal to bring to the design subgroup (typically because proposals are immature, incomplete, and/or competing), and there is a strong candidate to chair the SG to ensure it operates productively." (SD-4, "Subgroup responsibility and polls")
m. **Fact.** "The design subgroups (EWG and LEWG) own selecting and recommending feature designs and ship vehicles (TS or IS)." (SD-4, "Subgroup responsibility and polls")
n. **Fact.** "Both positive and negative comments are welcome, but should always be constructive and respectful." (SD-4, "Subgroup responsibility and polls")
o. **Fact.** "If anyone has a concern that a particular new proposal should not take up extended subgroup time (e.g., is not well thought out, is in a direction that previously did not get interest or consensus, or is otherwise unlikely to progress in its current form), they should share their concern privately and in advance with the subgroup chair." (SD-4, "Subgroup responsibility and polls")
p. **Fact.** "The subgroup chair may choose (based on such requests or just on their own judgment) to expedite the proposal's initial handling to make sure a representative overview is heard but then determine whether it has enough subgroup interest before spending extended time exploring the details; for example, the chair might time-box initial discussion and then poll whether the subgroup is interested in discussing the proposal's details further." (SD-4, "Subgroup responsibility and polls")

q. **Fact.** "The proposal author is expected to participate in the design subgroup to present the proposal, answer questions, participate in design discussion, and implement any changes directed by the group, typically for subsequent revisions of the proposal to be considered at the next meeting. In rare cases, if the proposal author chooses not to implement a change directed by the group, the author should be prepared to defend that decision and bring new information to the next meeting as part of the paper revision; proposals that do not implement design subgroup direction harm their own chances of gaining consensus within the design group and/or in plenary." (SD-4, "Subgroup responsibility and polls")

r. **Fact.** "When a proposal's design achieves a general consensus of the design subgroup, as determined by the subgroup chair, for its design and ship vehicle, it is forwarded to the specification subgroups." (SD-4, "Subgroup responsibility and polls")

s. **Fact.** "The specification subgroups (CWG and LWG) own the recommendation of the “standardese” specification (aka “wording”) that implements the intended design." (SD-4, "Subgroup responsibility and polls")

t. **Fact.** "If the specification subgroup has design questions or suggestions, those questions are referred back to the design group." (SD-4, "Subgroup responsibility and polls")

u. **Fact.** "When a proposal's specification achieves a general consensus of the specification subgroup, as determined by the subgroup chair, that its specification correctly expresses the intended design, it is forwarded to the closing plenary (whole-WG21) session." (SD-4, "Subgroup responsibility and polls")

v. **Fact.** "Subgroup chairs prioritize first papers that are related to one or more of the following: Direction priorities recommended in" ... "work the committee is aiming to include in the next IS; and work that can make plenary progress at this meeting. After that they get to things on a best-effort basis as time allows." (SD-4, "Subgroup responsibility and polls"; the omitted words are a paper number)

44. **Fact.** SD-4's section "Closing plenary session responsibility and polls" contains these statements.

a. **Fact.** "Design subgroup chairs present the proposals whose designs were approved at this meeting, including to solicit all present to express any serious design concerns that may not have been discovered in subgroups so that these can be known and addressed before extensive wording effort is invested." (SD-4, "Closing plenary session responsibility and polls")

b. **Fact.** "The default is to ask if there is objection to unanimous consent. (Unanimous consent means that all participant positions are either Favor or Neutral, and that none are Against). Most polls pass by unanimous consent." (SD-4, "Closing plenary session responsibility and polls")

c. **Fact.** "If there is an objection to unanimous consent, the Convener takes a 3-way poll as described above. If there is significant opposition, the Convener will usually ask whether the Against votes are personal positions or national body positions, typically by asking the national body panel chairs who are present." (SD-4, "Closing plenary session responsibility and polls")

d. **Fact.** "The WG21 closing plenary session owns approval of adding a proposal and its specification wording to a working draft (TS or IS). The plenary session's responsibility is to decide whether

there is consensus, as determined by the Convener, to adopt each proposed change that is recommended by the design and specification subgroups." (SD-4, "Closing plenary session responsibility and polls")

e. **Fact.** "The plenary session does not actively change the design or entertain extended design discussion, which always happens in the design subgroups." (SD-4, "Closing plenary session responsibility and polls")

f. **Fact.** "However, the plenary session participants may ask questions like: “does this handle case X?” and “did you consider alternative Y?” For such questions, the meeting chair will ask the appropriate subgroup chair (who may invite comment from the proposal's author) to answer whether that item was considered and to summarize the result of the discussion and the reason for the decision of the subgroup." (SD-4, "Closing plenary session responsibility and polls")

45. **Fact.** SD-4's section "Technical Specifications (TS) and White Papers" contains these statements.

a. **Fact.** "A TS or white paper is an experimental branch. In ISO, a TS or white paper is for a feature that does not yet have consensus for an IS. A TS or white paper is a separately published “experimental branch” that can ship separately to gain experience — either with an experimental part of the design, or to get enough overall experience so that participants are comfortable that the overall design works as well as hoped. We usually do not put work into a TS or white paper unless WG21 likes the general direction the feature is exploring, and would entertain merging it into the IS in some form (possibly after significant further transformation) if it turns out to work out successfully; features we know we do not intend to put in the IS are not worth shipping in experimental form either." (SD-4, "Technical Specifications (TS) and White Papers")

b. **Fact.** Among the ways SD-4 says a TS or white paper differs from a feature in a draft standard, it "asks implementations/users to give feedback to identify shortcomings and puts them on notice to expect breaking changes (they typically should not rely on the feature in production code)." (SD-4, "Technical Specifications (TS) and White Papers")

c. **Fact.** SD-4 describes "pre-merge issues, where we do not have a consensus to merge into the IS working draft until this specific set of issues is first addressed." and says "Issues that people and national bodies were willing to hold in abeyance with provisional resolutions for the TS or white paper are expected to have to be addressed, possibly pre-merge; however, people are encouraged not to rehash decisions that already had consensus when the TS or white paper was started unless new information has emerged." (SD-4, "Technical Specifications (TS) and White Papers")

d. **Fact.** Of the pre-merge issues, SD-4 says "We encourage all participants to keep the second set as small as possible, for real make-or-break concerns only, so that we can merge a TS or white paper early to permit better “integration testing” as we refine it further in the full context of the IS, and permit other parts of the standard to adapt to the new feature." (SD-4, "Technical Specifications (TS) and White Papers")

e. **Fact.** "TS vs white paper: They are functionally the same, but a white paper has a very-lightweight ISO process and is now recommended by ISO and SC22." (SD-4, "Technical Specifications (TS) and White Papers")

f. **Fact.** "A white paper is adopted within our immediate parent committee SC22; it has no prescribed time schedule, no prescribed format, no ISO editing, and when technically complete is sent by WG21 to SC22 to be approved by a simple vote by SC22 national bodies to adopt the draft as a white paper." (SD-4, "Technical Specifications (TS) and White Papers")

g. **Fact.** "Once a feature is in an IS, it is essentially cast in stone and much more difficult to update with breaking changes." (SD-4, "Technical Specifications (TS) and White Papers")

h. **Fact.** "The TS or white paper should list known specific questions on which feedback is desired, including:" "What are the exit criteria (especially, the list of must-address controversial issues) before we can consider merging this TS or white paper into the IS?" (SD-4, "Technical Specifications (TS) and White Papers")

46. **Fact.** SD-4's section "Working draft handling" contains these statements.

a. **Fact.** "In the opening plenary session of each meeting, we routinely adopt the updated working drafts generated after the previous meeting as described in the next paragraph. During the meeting, the wording for all new proposed changes is written against these adopted drafts as the meeting's baseline." (SD-4, "Working draft handling")

b. **Fact.** "In the closing plenary session of each meeting, we have a series of polls each asking to approve applying changes specified in a given wording paper or issues list entry to update a specific working draft." (SD-4, "Working draft handling")

c. **Fact.** "After the meeting, each project editor produces an updated candidate working draft that applies all the changes approved for that draft, and generates an editor's report summarizing those changes and any other comments that arose while applying and merging them." "If a working draft is also ready to be sent out for ISO ballot, and there were changes approved to the draft, then the poll to direct the convener to send the document out for ballot also includes naming an editing committee to verify that all approved changes were applied correctly by the project editor before the document is transmitted for ballot." (SD-4, "Working draft handling")

47. **Fact.** SD-4's section "Ballots and comments" contains these statements.

a. **Fact.** "National bodies are expected to send experts to participate in person in WG21 if they want to affect the direction of the standard." (SD-4, "Ballots and comments")

b. **Fact.** "A Disapproval (No) vote should be used only to express that the national body does not want the work (TS or IS) to progress at all in any form — to block further progress on all parts of the work item." (SD-4, "Ballots and comments")

c. **Fact.** "Therefore, No should mean “No, we don't want this TS or IS at all,” because sufficient No votes have that effect — most ballots fail if more than 1/3 of NBs vote No, so voting No only to try to give the NB's comments a higher priority is ineffective (we consider all NB feedback at all times, not just on ballots, and regardless of the Yes/No vote on a ballot) and counterproductive (could actually cause the whole project to fail)." (SD-4, "Ballots and comments")

d. **Fact.** "Note: Ballot comments are treated by ISO, and therefore by WG21, as national body positions." (SD-4, "Ballots and comments")

e. **Fact.** "All editorial and technical comments on the content of that document are in scope. A comment is expected to contain a proposed resolution with sufficient detail to be actionable; if more detail is required than fits easily in the comment form, referencing or attaching a paper is encouraged." (SD-4, "Ballots and comments")
f. **Fact.** "A ballot comment that requests adding an additional feature that is not already in the document is out of scope. New feature proposals are encouraged, but the correct way to bring them to WG21 is by presenting a proposal paper and working to build consensus to add the feature." (SD-4, "Ballots and comments")
g. **Fact.** "A ballot comment that requests a change that was already considered and decided otherwise at a WG21 meeting, and comes from a national body that was present at the meeting and had an opportunity to have their objections be heard and considered, is out of harmony with the ISO Code of Conduct's commitment to 'accept group decisions.' Once the WG has consensus to send a document for ballot, to repeat as an NB comment an objection that previously failed to carry the day is actually making, not a new technical objection, but an objection to the consensus of the WG." "Note: There are two exceptions:" (SD-4, "Ballots and comments")
h. **Fact.** The first exception: "If sufficient new information came to light since the last discussion that it is reasonable to believe enough people could change their opinions to affect the previous consensus. That information, of course, requires an on-time paper for the ballot resolution meeting, and ideally that paper would be available early and referenced directly in the ballot comment." (SD-4, "Ballots and comments")
i. **Fact.** The second exception: "If the national body really strongly objects to a specific item so much that they are willing to vote No on progressing the whole TS or IS that it's in, and have informed the committee all along of the strength of their feeling (see “Escalation…” below). If WG21 were to ballot a document over those objections, they would know that a No With Comments could happen; but the No vote, and negative comments, would be regarding a previously discussed and decided matter, they should never arise for the first time as a surprise No vote on a ballot." (SD-4, "Ballots and comments")
j. **Fact.** "In the case of a successful final approval ballot, where after ballot comment resolution the document will be sent directly to ISO for publication without another ballot, any design change made between the ballot and publication will be expected to have near-unanimous consent in subgroups and in plenary." (SD-4, "Ballots and comments")

48. **Fact.** SD-4's section "How to report a potential technical concern, at any level of severity" contains these statements.
    a. **Fact.** "Describe the problem, and ask whether it was considered. If the concern is about a feature proposal (whether already adopted or still underway) include the proposal authors; as the feature's designers they will be among the best experts about the issue." (SD-4, "How to report a potential technical concern, at any level of severity")
    b. **Fact.** "As early as possible, inform the EWG and LEWG chairs and the proposal's authors: They need to be aware of your concern, are trained in triaging and prioritizing issues and concerns, and

can offer calm advice on how to proceed." "As early as possible, inform the subgroup: preferably during the discussion in a meeting, even more preferably before a meeting on a reflector." (SD-4, "How to report a potential technical concern, at any level of severity")

49. **Fact.** SD-4's section "Escalating a serious “cannot live with” objection, or significant new information" contains these statements.
    a. **Fact.** "As early as possible, make the EWG and LEWG chairs aware of your concern. If it is a subgroup decision made during the week and that will be brought forward to plenary that week, you should proactively raise it during the week:" "On the committee email lists: Ideally the issue should be raised on the relevant subgroup's email list immediately following the subgroup discussion to which the objection appertains, but in no case any later than the deadline of 5pm the evening before the closing plenary session (and posting much sooner than that is strongly encouraged, ideally right after the subgroup decision they feel is wrong, is strongly encouraged)." (SD-4, "Escalating a serious “cannot live with” objection, or significant new information")
    b. **Fact.** "Within your own national body: This is so that the national body chair has opportunity to caucus with their members and determine whether the position may be a national position (normally the default is no, not a national position, unless it was discussed before or during the meeting to determine that it is a national position)." (SD-4, "Escalating a serious “cannot live with” objection, or significant new information")
    c. **Fact.** "Objections that are not so escalated, but are raised or re-raised in plenary, should not be given weight in plenary." "(In the rare case of new information that is found 'just the night before' plenary, it has still passed the cutoff for plenary participants to be able to give it informed and prepared consideration in time for plenary, and it is a race condition that we linearize by saying it has failed to be raised before the cutoff, and should be handled per the next bullet including with a followup paper.)" (SD-4, "Escalating a serious “cannot live with” objection, or significant new information")
    d. **Fact.** "If it is new information found after the deadline (including between meetings), they should raise it:" "On the committee email lists." "As a national body ballot comment. If a participant does not raise the concern within their national body, and succeed in convincing enough people in their caucus to make it a national body position (by whatever internal rules that particular national body panel uses to determine such), it won't become a national position." (SD-4, "Escalating a serious “cannot live with” objection, or significant new information")
    e. **Fact.** "In both cases, it should be raised with a discussion paper containing detailed information (at least a draft paper if a final one is not yet available)." (SD-4, "Escalating a serious “cannot live with” objection, or significant new information")
50. **Fact.** SD-4's section "Other topics" contains these statements.
    a. **Fact.** "The following materials are not publicly available and are always password-protected:" "Documents on which ISO asserts copyright, notably the final text of a TS/IS." "Meeting records of subgroup discussion, meeting wikis, and non-public committee email lists (aka reflectors), which often include personal positions and discussion. It is not allowed to quote from these publicly (e.g.,

in papers and blog posts) except that the following are allowed: (a) quoting straw poll questions and numeric results; and (b) quoting words or positions attributed to a specific person with that person's prior consent." (SD-4, "Other topics")

b. **Fact.** "All meetings must be minuted. This includes telecons as well as in-person meetings." (SD-4, "Other topics")

51. **Fact.** ISO/IEC Directives Part 1, clauses 0.1 and 0.7, contain these statements.

a. **Fact.** "This document replaces the ISO/IEC Directives, Part 1:2024, including the ISO, IEC and JTC 1 Supplements." "For the first time in the history of ISO and IEC, a single common set of procedures for the technical work is applicable across ISO and IEC." (ISO/IEC Directives Part 1, 0.1)

b. **Fact.** "These procedures take into account the total cost of the operation. The concept of “total cost” includes direct expenditure by National Bodies, expenditure by the offices in Geneva (funded mainly by the dues of National Bodies), travel costs and the value of the time spent by Experts in working groups and committees at both national and international level." (ISO/IEC Directives Part 1, 0.7)

c. **Fact.** "Opportunities for remote participation at face-to-face meetings should be provided to the extent possible (see Clause 5)." (ISO/IEC Directives Part 1, 0.7)

d. **Fact.** "Consensus, which requires the resolution of substantial objections, is an essential procedural principle and a necessary condition for the preparation of International Standards that will be accepted and widely used. Although it is necessary for the technical work to progress speedily, sufficient time is required before the approval stage for the discussion, negotiation and resolution of significant technical disagreements." (ISO/IEC Directives Part 1, 0.7)

e. **Fact.** "Similarly, to avoid re-discussion, National Bodies have the responsibility to ensure that their technical standpoint is established, taking account of all interests concerned at the national level, and that this standpoint is made clear at an early stage of the work rather than, for example, at the final (approval) stage." (ISO/IEC Directives Part 1, 0.7)

f. **Fact.** "National Bodies are encouraged to abstain if there is a lack of national consensus or Expert input on a specific ballot." (ISO/IEC Directives Part 1, 0.7)

52. **Fact.** ISO/IEC Directives Part 1, clauses 1.7, 1.8.2, and 1.12.1 to 1.12.3, contain these statements about committees and working groups.

a. **Fact.** "Nevertheless, all National Bodies, irrespective of their status within a committee, have the right to vote on enquiry drafts (see 2.6) and on final draft International Standards (see 2.7)." (ISO/IEC Directives Part 1, 1.7)

b. **Fact.** Among the duties ISO/IEC Directives Part 1 gives the Chair of a committee (for WG21, the chair of SC22): "d) ensure at meetings that all points of view expressed are adequately summarized so that they are understood by all present;" and "e) ensure at meetings that all decisions are clearly formulated and made available in written form by the Secretary/Committee Manager for confirmation during the meeting;" (ISO/IEC Directives Part 1, 1.8.2)

c. **Fact.** "A WG operates by consensus, reports and gives recommendations, if any, to its parent committee through a WG Convenor appointed by the parent committee. The Convenor shall act in

a purely international capacity, divesting themselves of a national position." (ISO/IEC Directives Part 1, 1.12.1)

d. **Fact.** "WG Convenors shall be appointed by the committee for up to three-year terms. Such appointments shall be confirmed by their National Body (or liaison). The Convenor may be reappointed for additional terms of up to three years. There is no limit to the number of terms. Prior to the expiry of the term, the Secretary/Committee Manager shall issue a six-week minimum call to identify any candidates who wish to be considered for the position. If multiple candidates are nominated, the committee shall appoint a Convenor from among the candidates." (ISO/IEC Directives Part 1, 1.12.1)
e. **Fact.** "A WG comprises a number of Experts individually appointed by the P-members, category A liaison and Liaison Representatives from committees in liaison (see 1.15 and 1.16) and category C organizations (see 1.17)." (ISO/IEC Directives Part 1, 1.12.1)
f. **Fact.** "The Experts appointed by a P-member or category A liaison act in a personal capacity and not as the official representative of them. The Liaison Representatives appointed by a category C liaison or committees in liaison act as official representatives of the organizations by which they have been appointed." (ISO/IEC Directives Part 1, 1.12.1)
g. **Fact.** "It is recommended that the Experts and Liaison Representatives keep close contact with that P-member or liaison to inform them about the progress of the work and of the various opinions in the WG at the earliest possible stage." (ISO/IEC Directives Part 1, 1.12.1)
h. **Fact.** "It is recommended that WGs be reasonably limited in size. The committee may therefore decide upon the maximum number of Experts appointed by each P-member and liaison organization." (ISO/IEC Directives Part 1, 1.12.1)
i. **Fact.** "Experts not registered to a WG in the ISO GD or the IEC EMS, as appropriate, shall not participate in its work." (ISO/IEC Directives Part 1, 1.12.2)
j. **Fact.** "Convenors may invite a specific guest to participate in up to two meetings to provide relevant expertise to the WG and shall notify the guest's National Body of the invitation ahead of the meeting via the Office of the CEO (copying the Secretary/Committee Manager of the committee)." (ISO/IEC Directives Part 1, 1.12.2)
k. **Fact.** "Convenors should review the participation in the WG at regular intervals to identify persistently inactive P-member Experts." "In the absence of satisfactory responses from the relevant P-members, the inactive P-member Experts shall be removed by the Office of the CEO." (ISO/IEC Directives Part 1, 1.12.3)

53. **Fact.** ISO/IEC Directives Part 1, clauses 2.5.6, 2.5.7, 2.5.9, and 2.5.10, contain these statements about consensus and the committee stage.

a. **Fact.** "Consideration of successive drafts shall continue until a consensus of the P-members as judged by the Chair has been obtained, or a decision to abandon or defer the project has been made." (ISO/IEC Directives Part 1, 2.5.6) "The decision to circulate an enquiry draft (see 2.6.1) shall be taken by the Chair of the committee on the basis of the consensus among the P-members." "In case of doubt concerning consensus on a revised CD, approval by a two-thirds majority of the

P-members voting may be deemed to be sufficient for the revised CD to be accepted for registration as an enquiry draft; however, every attempt shall be made to resolve negative votes." (ISO/IEC Directives Part 1, 2.5.7)

b. **Fact.** "It is the responsibility of the Chair of the committee, in consultation with the Secretary/Committee Manager of the committee and, if necessary, the Project Leader, to judge whether there is consensus bearing in mind the definition of consensus given in ISO/IEC Guide 2:2004 [11]." (ISO/IEC Directives Part 1, 2.5.7)
c. **Fact.** "However, “sustained oppositions” are views expressed at minuted meetings of the committee, working group (WG) or other groups (e.g. advisory groups) and which are maintained by an important part of the concerned interest and which are incompatible with the committee consensus." (ISO/IEC Directives Part 1, 2.5.7)
d. **Fact.** "The notion of “concerned interest(s)” will vary depending on the dynamics of the committee and shall therefore be determined by the committee leadership on a case-by-case" ... "basis." (ISO/IEC Directives Part 1, 2.5.7; the omitted text is a page number and copyright line)
e. **Fact.** "The concept of sustained opposition is not applicable in the context of National Body votes on enquiry or approval ballots since these are subject to the applicable voting rules." (ISO/IEC Directives Part 1, 2.5.7)
f. **Fact.** "Those expressing sustained opposition have a right to be heard, and the following approach is recommended when a sustained opposition is declared." "The leadership shall first assess whether the opposition can be considered a “sustained opposition”, i.e. whether it has been sustained by an important part of the concerned interest. If this is not the case, the leadership will register the opposition (i.e. in the report/minutes, records, etc.) and continue to lead the work on the document." (ISO/IEC Directives Part 1, 2.5.7)
g. **Fact.** "If the leadership determines that there is a sustained opposition, it shall try and resolve it in good faith. However, a sustained opposition is not akin to a right to veto. The obligation to address the sustained oppositions does not imply an obligation to resolve them successfully." (ISO/IEC Directives Part 1, 2.5.7)
h. **Fact.** "The responsibility for assessing whether or not a consensus has been reached rests entirely with the leadership. This includes assessing whether there is sustained opposition or whether any sustained opposition can be resolved without compromising the existing level of consensus on the rest of the document. In such cases, the leadership will register the opposition and continue the work." (ISO/IEC Directives Part 1, 2.5.7)
i. **Fact.** "Those parties with sustained oppositions may avail themselves of appeals mechanisms as detailed in Clause 6." (ISO/IEC Directives Part 1, 2.5.7)

j. **Fact.** "The committee stage ends when all technical issues have been resolved, and a CD is accepted for circulation as an enquiry draft and is registered by the Office of the CEO." (ISO/IEC Directives Part 1, 2.5.9) "If the technical issues with an International Standard project cannot all be resolved within the appropriate time limits, committees may consider publishing an intermediate document in the form of a Technical Specification (see 3.1) pending agreement on an International Standard." (ISO/IEC Directives Part 1, 2.5.10)

54. **Fact.** ISO/IEC Directives Part 1, clauses 2.6.3 to 2.6.5, 2.7.2 to 2.7.7, 3.1.2, and 3.1.3, contain these statements about ballots and Technical Specifications.
    a. **Fact.** "An enquiry draft is approved if" "a) a two-thirds majority of the votes cast by the P-members are in favour, and" "b) not more than one-quarter of the total number of votes cast are negative." "Abstentions are excluded when the votes are counted, as well as negative votes not accompanied by any comments of a technical nature." (ISO/IEC Directives Part 1, 2.6.3)
    b. **Fact.** "If a National Body finds a final draft International Standard unacceptable, it shall vote negatively and state the reasons by submitting comments of a technical nature. It shall not cast a positive vote that is conditional on the acceptance of modifications." (ISO/IEC Directives Part 1, 2.7.2)
    c. **Fact.** After an enquiry ballot, one course of action open to the Chair of the committee (for WG21, the chair of SC22) is "a) when the approval criteria of 2.6.3 are met, and no technical changes are to be included, to proceed directly to publication (see 2.8);" (ISO/IEC Directives Part 1, 2.6.4) At the approval stage, "The Secretary/Committee Manager has the responsibility to bring any errors that may have been introduced in the preparation of the draft to the attention of the Office of the CEO by the end of the voting period; further editorial or technical changes are not accepted at this stage." (ISO/IEC Directives Part 1, 2.7.4) "Technical changes to an approved FDIS are not allowed." (ISO/IEC Directives Part 1, 2.7.5)
    d. **Fact.** "If the final draft International Standard is not approved in accordance with the conditions in 2.7.3, the document shall be referred back to the committee concerned for reconsideration in light of the technical reasons submitted in support of the negative votes." "The committee shall decide to:" "resubmit a modified draft as a committee draft, enquiry draft or final draft International Standard; or" "publish a Technical Specification (see 3.1); or" "cancel the project." (ISO/IEC Directives Part 1, 2.7.7)
    e. **Fact.** "All comments received will be retained for the next review and will be recorded on the voting form as “noted for future consideration”." (ISO/IEC Directives Part 1, 2.7.5)
    f. **Fact.** "Not later than 12 weeks after the end of the voting period, a full report shall be prepared by the secretariat of the committee and circulated by the Office of the CEO to the National Bodies. The report shall" "a) show the result of the voting," "b) state the decision of the Chair of the committee," "c) reproduce the text of the comments received, and" "d) include the observations of the secretariat of the committee on each of the comments submitted." (ISO/IEC Directives Part 1, 2.6.5)
    g. **Fact.** "Every attempt shall be made to resolve negative votes." "If, within eight weeks from the date of dispatch, two or more P-members disagree with the decision of the Chair, the draft shall be

discussed at a meeting (see Clause 5)." "Committees are required to respond to all comments received." (ISO/IEC Directives Part 1, 2.6.5)

h. **Fact.** "A final draft International Standard having been circulated for voting, is approved if" "a) a two-thirds majority of the votes cast by the P-members are positive, and" "b) not more than one-quarter of the total number of votes cast are negative." "Abstentions are excluded when the votes are counted, as well as negative votes not accompanied by any comments of a technical nature." (ISO/IEC Directives Part 1, 2.7.3)

i. **Fact.** "The decision to publish the resulting document as a Technical Specification shall require a two-thirds majority vote of the P-members voting. If technical changes are required following the ballot, a subsequent ballot is required to approve the revised draft." (ISO/IEC Directives Part 1, 3.1.2) "Competing Technical Specifications offering different technical solutions are possible, provided that they do not conflict with existing International Standards." (ISO/IEC Directives Part 1, 3.1.3)

55. **Fact.** ISO/IEC Directives Part 1, clauses 5.3.1, 5.5, and 5.7, contain these statements about meetings.

a. **Fact.** "Participants shall not comment publicly on the discussions, positions, or contributions made by others during meetings." "Only the meeting leadership may speak publicly on the consensus reached, and only in the context of an official press activity linked to the meeting." (ISO/IEC Directives Part 1, 5.7)

b. **Fact.** "Meetings hosts shall not charge participants any sort of participation fee, nor require accommodation at specific hotels or hotel rates for any meetings." (ISO/IEC Directives Part 1, 5.3.1)

c. **Fact.** "The Secretary/Committee Manager or Project Leader/Convenor shall ensure that meeting outcomes are properly documented and shared. Decisions shall be distributed within 48 hours after the meeting, and the meeting report/minutes shall be distributed within four weeks." "If no comments are received within four weeks after distribution, the meeting report/minutes is considered approved. If comments are received, they shall be addressed by the Secretary/Committee Manager or Convenor. If objections are received, the meeting report/minutes remains unconfirmed until the next meeting, and any approved changes are recorded in the meeting report/minutes of that next meeting." (ISO/IEC Directives Part 1, 5.5)

56. **Fact.** ISO/IEC Directives Part 1, clause 6, contains these statements about appeals.

a. **Fact.** "National Bodies have the right of appeal" "a) to the parent technical committee on a decision of a subcommittee;" "b) to the technical management board on a decision of a technical committee;" "c) to the council board on a decision of the technical management board," "within eight weeks of the decision in question." (ISO/IEC Directives Part 1, 6.1.1)

b. **Fact.** "A P-member of a committee may appeal against any action, or inaction, on the part of the committee, when the P-member considers that such action or inaction is" "a) not in accordance with" "the Statutes and Rules of Procedure [16], [17];" "the ISO/IEC Directives; or" "b) not in the best interests of international trade and commerce, or such public factors as safety, health or environment." (ISO/IEC Directives Part 1, 6.1.2)

c. **Fact.** "Matters under appeal may be either technical or administrative in nature." "Appeals on decisions concerning new work item proposals, committee drafts, enquiry drafts and final draft International Standards are only eligible for consideration if" "questions of principle are involved, or" "the contents of a draft may be detrimental to the reputation of ISO or IEC." (ISO/IEC Directives Part 1, 6.1.3)

d. **Fact.** "When an appeal is against a decision relating to work in progress, the work shall be continued up to and including the approval stage (see 2.7)." (ISO/IEC Directives Part 1, 6.5)

57. **Fact.** INCITS's Organization, Policies and Procedures contain these statements on how INCITS counts votes and forms United States positions.

a. **Fact.** "There shall be only one voting membership per organization (see Section 3.9.2, Limits on Voting Membership). An alternate representative may vote only if the principal representative fails to vote. Voting by proxy shall not be permitted." (INCITS Organization, Policies and Procedures, 5.2)

b. **Fact.** The glossary defines "Organization (For the purposes of membership and voting)" as "Corporations, Federal or military departments or agencies, partnerships or associations, self-employed or self-financed individuals, or any other legal or commercial entities." (INCITS Organization, Policies and Procedures, 8.2)

c. **Fact.** "Note: On TCs, TGs, and SGs, the U.S. Department of Defense (DoD) is allowed a maximum of four votes from separate and unrelated entities such as the Department of Army, Department of Navy, Department of Air Force, Defense Logistics Agency, Defense Mapping Agency, etc. This exception is due to the unique nature of the DoD and is, therefore, not considered a precedent for any further exceptions." "In order to be in the voting member category, consultant organizations shall declare that their participation is not being funded by any organization already having voting membership or by an organization that is not eligible for membership." For standards developing organizations and consortia: "There shall be only one voting membership for each separate organization. For these entities, their principal and alternate representatives may be employed by other organizations that have voting memberships." (INCITS Organization, Policies and Procedures, 3.9.2)

d. **Fact.** "The appropriate U.S. TAG shall submit a recommendation on the U.S. voting position to the Executive Board. Once the U.S. position is approved by the Executive Board, it shall be submitted to ANSI for transmission to JTC 1 or ISO." (INCITS Organization, Policies and Procedures, 7.11.2) "The INCITS Executive Board is the U.S. TAG to JTC 1." (INCITS Organization, Policies and Procedures, 8.2)

e. **Fact.** "When a new issue or question comes before a Plenary, or an existing issue or question is modified during the course of the Plenary, the U.S. delegation may determine the official U.S. position by consensus of the U.S. caucus as determined by the HoD." (INCITS Organization, Policies and Procedures, 7.17.1.4)

58. **Fact.** Each item below states an absence that a search of the documents and clauses it names confirms; the edition of the Directives these items search replaces the JTC 1 Supplement, and the proof does not search the other documents SD-4 lists as covering material it does not repeat.
    a. **Fact.** SD-4 contains no requirement that the record of a poll show how an objection was answered or why a consensus call was made (SD-4, whole document; searched: record, minute, reason, answer, tally, objection, why).
    b. **Fact.** SD-4 names no step in which a chair asks voters of any position other than Against for their reasons (SD-4, whole document; searched: reason, rationale, articulate, explain, justif, why).
    c. **Fact.** None of ISO/IEC Directives Part 1 clauses 1.8.2, 2.5.7, and 5.5, which govern how decisions, minutes, and oppositions are recorded, requires a record of why consensus was judged to exist (ISO/IEC Directives Part 1, 1.8.2, 2.5.7, 5.5; searched: reason, rationale, justif, why).
    d. **Fact.** SD-4 uses the words "sustained" and "concerned interests" only in its quotation of the definition of consensus, names no step for a sustained opposition as such, gives no criteria for an important part of the concerned interests, and has no step for registering an opposition (SD-4, whole document; searched: sustained, concerned interest, important part, register, record, minute, assess, address, resolve, opposition, objection, Against).
    e. **Fact.** Neither SD-4 nor any ISO/IEC Directives Part 1 clause listed here limits the number of participants whose costs one sponsor pays (SD-4, whole document; ISO/IEC Directives Part 1, 0.7, 1.7, 1.8.2, 1.12.1 to 1.12.3, 2.1.6, 2.1.7, 2.5.6 to 2.5.10, 2.6.2 to 2.6.5, 2.7.2 to 2.7.7, 3.1, 5.3.1, 5.4 to 5.7, 6.1 to 6.5; searched: sponsor, employer, company, organization).
    f. **Fact.** SD-4 assigns no one the role of ruling which side of a technical dispute is correct or of judging a design's quality (SD-4, whole document; searched: rule, ruling, decide, arbiter, correct, authority, quality, judge, evaluat, assess, merit).
    g. **Fact.** SD-4 contains no requirement that a concern raised privately with a chair be recorded (SD-4, whole document; searched: privately, record, minute, concern).
    h. **Fact.** Beyond showing the design alternatives considered and bringing new information when declining a directed change, SD-4 asks no proposal paper to state the objections raised against its proposal; it asks a TS or white paper for its exit criteria, "especially, the list of must-address controversial issues" (SD-4, whole document; searched: objection, concern, answer, address, rebut, respon, rationale, revision, feedback, oppos, disagree).
    i. **Fact.** SD-4 asks no proposal paper to state its design's drawbacks, and of the searched terms it uses only "cost", for travel costs (SD-4, whole document; searched: trade-off, tradeoff, trade off, cost, drawback, downside, disadvantage, limitation, weakness).
    j. **Fact.** SD-4 asks no author to state who pays their costs or what interests they or a sponsor have in a proposal (SD-4, whole document; searched: disclos, declar, affiliat, interest, employ, sponsor, fund, compan, behalf).
    k. **Fact.** SD-4 names no step in which anyone checks whether designs adopted in separate papers work together (SD-4, whole document; searched: interact, integrat, coheren, consisten, together, harmon, conflict, fits).

l. **Fact.** SD-4 asks no paper that proposes part of a design to state what later parts will require of it, and names no step for reviewing such parts together (SD-4, whole document; searched: part, parts, later, future, rule out, preclud, foreclos, extension, extend, subsequent, follow-on, builds on, depend, together).

In plain terms: the proof starts from assumptions about cost, time, preference, standing, learning, and memory, from defined terms that include two kinds of author, and from the rules as SD-4 and the ISO/IEC Directives state them, in the edition of the Directives that replaces the JTC 1 Supplement. For the United States position it also uses INCITS's rules for counting votes.

2. Who decides, and with what moves

59. **Proposition.** The participants present decide a design's content: chairs call consensus on their votes, counted under the 2:1 norm, in the design rooms and the study groups feeding them. SD-4 assigns no one to judge a design's quality, wording rooms check only that wording expresses the design, and national bodies ballot only on whole documents. *[proved by 59a, 59b, 59c, 59d, 59e] [from 43b, 43f, 58f]*
    a. **Lemma.** The design rooms select designs, and a design room sends one to the wording rooms once its chair determines that it has the room's general consensus. *[from 43m, 43r, 24]*
    b. **Lemma.** A study group produces the initial proposal a design room starts from, and its chair may poll alternatives side by side to see which has most consensus. *[from 43k, 43c, 43d, 24]*
    c. **Lemma.** A wording room sends wording to plenary once its chair determines a consensus that the wording correctly expresses the intended design, and sends design questions back to the design room. *[from 43s, 43u, 43t, 24]*
    d. **Lemma.** Plenary decides, by the convenor's consensus call, whether to adopt each change the design and wording rooms recommend, and leaves design to the design rooms. *[from 44d, 44e, 24]*
    e. **Lemma.** A national body's ballot vote counts for or against a whole draft. SD-4 reserves No for blocking every part of the work, and on a final draft ISO bars a Yes conditional on changes. *[from 54a, 54h, 47b, 54b, 25]*
60. **Proposition.** Each person in a subgroup's room casts one vote, guests a convenor invites included, while no user votes in any WG21 poll or in the ballots on its drafts. *[from 43b, 52j, 52a, 54h, 18, 23, 25]*
61. **Proposition.** The players are result-seeking authors, design-bound authors, regulars, study group chairs, design room chairs, the convenor, national bodies, sponsors, and users. The proof takes as given the choices of wording room chairs, guests, participants who are not regulars, and the direction group, and the convenor's appointments, study groups, and guest invitations. *[from 59, 60, 34, 21, 37, 38, 19, 24, 25, 22, 23]*
62. **Proposition.** Each player has these moves under the rules. *[proved by 62a, 62b, 62c, 62d, 62e, 62f, 62g]*
    a. **Lemma.** An author can submit papers, including papers that each propose part of a design, and present them or have a qualified and prepared presenter do so. An author can also revise them and ask the chair for a poll or the design room for a ship vehicle, which the chair and the design rooms decide. *[proved by 62a.i, 62a.ii] [from 42a, 42h, 42j, 43c, 43m, 21]*

   i. A part of a design proposed on its own is a design addition, which SD-4 requires to have a paper. *[from 42a]*
   ii. Like every paper in the mailings since the previous meeting, one proposing part of a design is on the agenda, and SD-4 names no step for reviewing the parts together. *[from 42d, 58l]*

b. **Lemma.** A regular can attend or not, study a paper or not, speak, and vote Favor, Neutral, or Against, or not at all. A regular can also write a paper, raise a concern privately with a chair, and ask their national body to take a national position. *[from 1, 6, 43n, 43b, 43g, 42a, 42g, 43o, 49b, 19]*

c. **Lemma.** A chair can choose the polls, order the agenda by SD-4's priorities, time-box a new proposal and poll whether to go on, re-poll after discussion, ask Against voters for reasons, and allow limited exploration of alternatives. A design room or wording room chair also makes the consensus call that forwards a proposal, a design room chair can delay a proposal one meeting for a rival without a paper, and a study group chair polls which design leaves the group as its initial proposal. *[from 43c, 43v, 43i, 42d, 43p, 43f, 43h, 42l, 43r, 43u, 42k, 43k, 24]*

d. **Lemma.** The convenor can appoint chairs, form study groups, invite a guest for up to two meetings, and make plenary consensus calls. In plenary the convenor can ask whether Against votes are personal or national body positions. *[from 43a, 43l, 52j, 44d, 44b, 44c]*

e. **Lemma.** A national body can appoint experts, vote Yes or No or abstain on ballots, comment, state a national position in plenary, and appeal. It can also object in SC22 and JTC 1 to action taken on a topic not on the agenda. *[from 52e, 54h, 52a, 51f, 47d, 44c, 42e, 56a]*

f. **Lemma.** A sponsor can pay for participation and choose whom and how many it pays, and a category C liaison's representatives act for it. Unless it is an ISO member body or a liaison organization, a sponsor has no vote of its own in WG21's polls or in the ballots on WG21's drafts. *[from 22, 2, 58e, 43b, 52e, 52f, 54h, 52a, 25]*

g. **Lemma.** A user casts no vote in WG21's polls or ballots but can give feedback on a TS or white paper or write a paper a qualified and prepared participant presents. A user can also raise their interests with a national body, which ISO asks to take account of all interests concerned at the national level, or become a participant. *[proved by 62g.i] [from 60, 45b, 42h, 51e, 23]*
   i. A person becomes a participant once a national body or category A liaison appoints them as an expert, they are registered, and they attend a WG21 meeting. *[from 18, 52e, 52i, 25]*

In plain terms: the participants present decide a design's content through consensus calls on their votes under the 2:1 norm, and SD-4 assigns no one to judge a design's quality. No user casts a vote in WG21's polls or ballots, and a sponsor has no vote of its own there unless it is an ISO member body or a liaison organization. The players are result-seeking and design-bound authors, regulars, study group and design room chairs, the convenor, national bodies, sponsors, and users, each with the moves the rules provide.

3. What a vote costs

63. **Lemma.** Under the 2:1 norm each Against vote raises by two the Favor votes a proposal needs, while Neutral votes and abstentions leave the count unchanged. One Against vote changes the result only at

one or two votes above the line, and at least a third of the Favor-plus-Against votes cast Against keeps the norm unmet. *[proved by 63a, 63b, 63c, 63d]*

a. **Lemma.** Under the 2:1 norm, each Against vote raises by two the number of Favor votes a proposal needs. *[from 43f, 26]*
b. **Lemma.** Neutral votes and abstentions do not enter the 2:1 comparison, though SD-4 says a Neutral majority of those voting can be concerning. *[from 43f, 43g, 26]*
c. **Lemma.** One more Against vote changes whether the 2:1 norm is met only when the Favor votes exceed twice the other Against votes by one or two. *[proved by 63c.i, 63c.ii] [from 43f, 26]*
    i. Without that vote, the 2:1 norm is met exactly when the Favor votes exceed twice the other Against votes by one or more. *[from 43f, 26]*
    ii. With that vote, the norm is met exactly when the Favor votes exceed twice the other Against votes by three or more, two more than before. *[from 63a]*
d. **Lemma.** Any group casting at least one third of the Favor-plus-Against votes as Against prevents the 2:1 norm from being met. *[proved by 63d.i, 63d.ii] [from 43f, 26]*
    i. The 2:1 norm needs more than twice as many Favor votes as Against votes, that is, Favor votes making up more than two thirds of the Favor-plus-Against votes. *[from 43f, 26]*
    ii. A group casting at least a third of the Favor-plus-Against votes as Against leaves the Favor votes at most two thirds of them. *[from 26]*

64. **Proposition.** For a regular, an Against vote on a colleague's proposal costs more than a Neutral vote or none. It risks the colleague's later support no less, and only Against voters can be asked for reasons, which take time to prepare. *[from 5, 3, 4, 7, 43h, 58b, 19, 27]*
65. **Proposition.** A regular's doubt about a colleague's proposal enters the 2:1 count only if it is worth the extra cost of an Against vote. That vote changes the result only when the count sits one or two votes above the line, though it can also win changes to the design. *[proved by 65a, 65b] [from 63, 64, 4, 43h, 43q, 28, 29]*
    a. **Lemma.** When an Against vote is worth less to the regular than its extra cost, the regular ranks a Neutral vote or none above it. *[proved by 65a.i, 65a.ii] [from 64, 4, 29]*
        i. Switching from a Neutral vote or none to Against brings the regular what the Against vote is worth, at its extra cost in standing risked and time for reasons. *[from 64]*
        ii. When that worth is less than the extra cost, the switch leaves the regular worse off, so the regular ranks a Neutral vote or none above Against. *[from 65a.i, 4, 29]*
    b. **Lemma.** When an Against vote is worth more to the regular than its extra cost, the regular ranks it above a Neutral vote or none. *[proved by 65b.i] [from 64, 4, 29]*
        i. When the Against vote's worth exceeds its extra cost, switching to it from a Neutral vote or none leaves the regular better off, so the regular ranks it first. *[from 65a.i, 4, 29]*

Table 1. One regular's vote on a colleague's proposal that the regular doubts. Columns: what an Against vote is worth to the regular, compared with its added cost. Each cell: the regular's rank of the vote. 2 is best within each column.

	Worth below cost	Worth above cost
Neutral or abstain	**2**	1
Against	1	**2**

[from 65a, 65b]

66. **Proposition.** With many Favor votes cast, one objector's Against vote changes the result only if enough others also vote Against, so its worth to each objector depends on the others' votes. The same objectors can then form either of two settled patterns: just enough of them vote Against to block, or none does. *[proved by 66a, 66b, 66c] [from 63, 65, 64, 28, 34]*
    a. **Lemma.** With many Favor votes cast, an objector's Against vote leaves the result unchanged when no other objector votes Against, or when enough others already do to block. It changes the result when it is one of just enough Against votes to block, since without it the 2:1 norm would be met. *[from 63]*
    b. **Lemma.** None of the objectors voting Against is a settled pattern when an Against vote that leaves the result unchanged is worth less to each of them than its extra cost. *[from 66a, 65a, 34]*
    c. **Lemma.** In that case, just enough objectors voting Against to block is also a settled pattern when each of them values their vote above its extra cost. *[from 66a, 65b, 65a, 34]*
67. **Proposition.** A participant can argue for another design and still vote Neutral, which leaves the count unchanged and draws no request for reasons. *[from 43n, 63, 43h, 58b]*
68. **Proposition.** Two regulars with pending proposals each risk the other's support for their own by opposing the other's proposal. When each regular's doubt is worth less than that risk and the time for reasons, neither opposes, and both do better than if both opposed. *[proved by 68a, 68b] [from 5, 4, 64, 65, 63, 29]*
    a. **Lemma.** In the case the table below describes, each regular ranks refraining above opposing, whatever the other does. That holds because an Against vote on the other's proposal is then worth less to them than its extra cost. *[proved by 68a.i, 68a.ii] [from 65, 65a, 5]*
        i. In that case, an Against vote on the other's proposal is worth less to each regular than its extra cost in standing risked and time for reasons. *[from 64, 28]*
        ii. So each regular ranks a Neutral vote or none above Against, whatever the other does, and refraining includes those votes. *[from 68a.i, 65a]*
    b. **Lemma.** In that case, each regular ranks opposing while the other refrains above refraining while the other opposes, since the other's opposition costs them more than their own opposing does. *[proved by 68b.i, 68b.ii] [from 4, 29]*
        i. Opposing while the other refrains costs a regular their own opposing, and refraining while the other opposes costs them the other's opposition to their proposal. *[from 64, 63]*
        ii. In that case the other's opposition costs each regular more than their own opposing does, so each ranks opposing while the other refrains above the reverse. *[from 68b.i, 4, 29]*

Table 2. Two regulars, each with a proposal pending, neither of which lowers what the other values by more than the cost of opposing it, and for each of whom the other's opposition costs more than their own opposing does. Refrains means votes Favor, Neutral, or not at all. First number: A's rank. Second number: B's rank. 4 is best. † marks a Pareto-optimal cell.

	B refrains	B opposes
A refrains	**4, 4 †**	2, 3
A opposes	3, 2	1, 1

[from 68a, 68b]

In plain terms: under the 2:1 norm, each Against vote raises by two the Favor votes a proposal needs, Neutral votes and abstentions leave the count unchanged, and at least a third of the Favor-plus-Against votes cast Against keeps the norm unmet. For a regular, an Against vote on a colleague's proposal costs more than a Neutral vote or none, so a doubt enters the count only if it is worth that extra cost. Two regulars with pending proposals each risk the other's support for their own by opposing the other's proposal, and when each one's doubt is worth less than that risk and the time for reasons, neither opposes and both do better than if both opposed.

4. What the record keeps and how objections count

69. **Proposition.** A proposal can advance in WG21's rooms and plenary with objections unresolved, since it needs a consensus call, which the 2:1 norm normally supports, not every objector's persuasion. *[proved by 69a, 69b]*
    a. **Lemma.** Under ISO, the obligation to address a sustained opposition is no obligation to resolve it, and a sustained opposition is not a veto. Still, ISO's general principles say consensus requires the resolution of substantial objections. *[from 53g, 51d]*
    b. **Proposition.** A proposal can advance while discussed objections to it remain unresolved, because advancing requires a consensus call, which the 2:1 norm normally supports, and consensus need not imply unanimity. *[from 69a, 43r, 43u, 44d, 43f, 41a]*
70. **Proposition.** The record SD-4 and the searched ISO clauses require of a WG21 poll need not show whether an objection was answered or outvoted, or why consensus was called over it. *[proved by 70a, 70b, 70c] [from 58c]*
    a. **Lemma.** SD-4 requires minutes and lets poll questions and numeric results be quoted publicly, but requires no record of how an objection was answered or why a consensus call was made. Plenary may ask a subgroup chair to summarize the reason for a decision. *[from 50b, 50a, 58a, 44f]*

b. **Lemma.** ISO lets participants comment on or object to minutes before they are confirmed, and has the chair of SC22 see that points of view at SC22's meetings are summarized. It bars participants from commenting publicly on others' contributions. *[from 55c, 52b, 55a]*

c. **Lemma.** The chair of SC22 judges consensus at SC22's committee stage, and ISO 2.5.7 recommends steps for a sustained opposition, including registering it, to the leadership. It does not say whether a convenor applies them inside a working group, and under either reading SD-4 adds no step for assessing or registering an opposition in subgroup polls. *[proved by 70c.i] [from 53b, 53a, 53j, 53f, 53h, 53c, 58d, 25]*

i. Clause 2.5.7 of ISO/IEC Directives Part 1 does not mention a convenor (ISO/IEC Directives Part 1, 2.5.7; searched: convenor, convener).

71. **Proposition.** SD-4 adopts ISO's definition of consensus, which turns on sustained opposition, but names no step for judging whether an opposition is sustained or for registering one, and no criteria for an important part of the concerned interests. Whoever ISO's leadership is for a working group, SD-4's own steps are the 2:1 norm after discussion, reasons from Against voters on request, escalation, and plenary's 3-way poll and question about national positions. *[proved by 71a] [from 52c, 41a, 53c, 43f, 43h, 49a, 44b, 44c, 53f, 58d]*

a. **Fact.** "Consensus is defined as follows in the ISO/IEC Guide 2:2004, as quoted in the ISO/IEC Directives:" (SD-4, "Consensus")

72. **Lemma.** Under SD-4's definition, an objection prevents consensus only if it concerns a substantial issue, is maintained, and comes from an important part of the concerned interests. The leadership judges that part case by case, none of the three conditions asks whether the objection is right, and every view must still be heard. *[proved by 72a] [from 41a, 53c, 41b, 40f]*

a. **Lemma.** ISO has the committee leadership determine the concerned interests case by case and assess whether an opposition has been sustained by an important part of them. *[from 53d, 53f, 53h]*

73. **Proposition.** When an objection's Against votes are too few to stop the 2:1 norm, the cheapest response keeping it met at that poll is no change or a partial change. Finding out whether those objectors are right leaves that count unchanged, and SD-4 expects any change the room directs to be made or defended with new information. *[proved by 73a, 73b] [from 4, 11, 7, 63, 43q]*

a. **Lemma.** A change to a design costs its authors the time to write a revised paper, and revising the design throughout costs them more than a change confined to one part. *[from 42a, 7, 11]*

b. **Lemma.** Studying whether objectors are right takes time, while the 2:1 comparison counts only how many vote Favor and how many vote Against. *[from 7, 63]*

74. **Proposition.** SD-4 asks an Against voter to be ready to name the changes that would win them over, or to say that none would. Concerns stated as changes give authors items to grant one at a time, while opposition in any form offers none and weighs in the poll only as a vote. *[from 43h, 62a]*

75. **Proposition.** Of minutes, non-public email lists, meeting records, ballot comments, and papers, only a paper puts an objection's argument outside the password-protected materials without anyone else's help. Writing one takes time. *[proved by 75a, 75b, 75c, 75d] [from 70, 50a, 55c, 47d, 49d, 49e, 62b, 7]*

a. **Lemma.** SD-4 keeps meeting records, meeting wikis, and non-public email lists password-protected, and allows quoting from them publicly only poll questions, numeric results, and words a person consents to have attributed. *[from 50a]*
b. **Lemma.** Under ISO, the secretary or convenor sees that meeting outcomes are documented and addresses comments on the minutes, so what minutes keep of an objection's argument rests with that officer. *[from 55c]*
c. **Lemma.** A ballot comment counts as a national body position, and a participant's concern becomes one only if enough members of their national body's caucus agree. *[from 47d, 49d]*
d. **Lemma.** A regular can write a paper stating an objection's argument, and SD-4 lists no paper among the materials it keeps password-protected. *[from 62b, 50a]*

76. **Proposition.** A proposal's case lasts for decades in its required paper, while an unwritten reason survives only with those who heard it and, once they all leave, only by rediscovery. Outside the protected records, an objection's argument outlasts its hearers only in writing, such as the objector's paper or an author's paper showing the alternative or defending a declined change. *[proved by 76a, 76b, 76c] [from 42a, 13, 75, 70, 42c, 43q]*
    a. **Lemma.** Every significant proposal needs a paper, and a published paper stays available to later readers for decades. *[from 42a, 13]*
    b. **Lemma.** An unwritten reason reaches someone absent only through those present who remember it. Once all of them have stopped attending, a later participant can voice it only by rediscovering it. *[from 13]*
    c. **Lemma.** Once its hearers have left, an objection's argument survives outside the protected records only in writing. That writing includes the objector's own paper and an author's paper showing the alternatives considered or defending a declined change. *[proved by 76c.i, 76c.ii, 76c.iii] [from 13, 75, 62b, 42c, 43q]*
        i. Someone absent learns what was said only from writing or from those present who remember it, so once all the hearers have left, only writing remains. *[from 13]*
        ii. Outside the protected records, a paper is the writing that states an objection's argument without anyone else's help, and a regular can write one. *[from 75, 62b]*
        iii. An author's paper can also state it, since SD-4 asks a good paper to show the alternatives considered and an author declining a directed change to defend that. *[from 42c, 43q]*
77. **Proposition.** SD-4 describes an objector who presses on after a consensus call, without new information or allowed escalation, as not accepting a group decision, whether or not anyone answered it. It says the same of a national body comment repeating a decided objection, unless the comment brings new information or the body warned all along that it would vote No. *[from 41c, 40e, 41d, 41e, 47g, 47h, 47i]*
78. **Proposition.** SD-4 gives an objection weight at plenary only if its holder escalated it by 5pm the evening before, on the email lists and within their national body. There, by default, it is not a national position, and after the deadline even new information goes to the email lists or a ballot comment, with a discussion paper. *[from 49c, 49a, 49b, 49d, 49e]*

79. **Proposition.** No design is considered on merit alone: a proposal needs an on-time paper and a presenter, a rival needs a paper to exist, and objections need fixed forms and times. Participants learn these rules partly by attending, some only that way, and other things equal, whoever knows them has the better chance of adoption, whatever their design skill. *[proved by 79a, 79b, 79c, 79d] [from 42f, 42h, 42k, 43b, 78, 47h, 12, 62]*
    a. **Lemma.** A proposal is considered at a meeting only with an on-time paper and an author or another qualified and prepared presenter to present it. *[from 42f, 42h]*
    b. **Lemma.** A competing approach without a paper does not exist for SD-4 and does not block a proposal before the room. *[from 42k]*
    c. **Lemma.** An objection counts in a subgroup poll only as a vote cast in the room, and gets weight at plenary only if escalated by the deadline. On a ballot, new information behind a decided objection needs an on-time paper. *[from 43b, 78, 47h]*
    d. **Lemma.** Participants learn these rules partly by attending, some only that way, and other things equal, those who know more have a better chance of adoption, whatever their design skill. *[from 12, 62]*
80. **Proposition.** Other things equal, a newcomer, lacking what earlier meetings teach, spends more effort getting a paper considered than a returning participant, though SD-4 expects everyone to know its procedures. Where WG21 runs study groups that incubate early proposals, such as EWGI and LEWGI, a newcomer's early proposal can start there. *[from 12, 20, 39b, 43k, 24]*

In plain terms: a proposal can advance with objections unresolved, and the record SD-4 and the searched ISO clauses require of a poll need not show whether an objection was answered or why consensus was called over it. When objectors are too few to stop the 2:1 norm, the cheapest response that keeps it met at that poll is no change or a partial one, and outside the protected records an objection's argument outlasts its hearers only in writing. Objections count only in fixed forms and at fixed times, and other things equal, whoever knows these rules, some of which only attending teaches, has the better chance of adoption.

5. Momentum: incumbents and the cost of reversal

81. **Induction.** Each decision built on a design adds work that reversing the design would redo or discard, and more is built after each favorable poll. So the cost of reversing a design grows as it advances, whether or not the design is wrong. *[from 10, 4, 33, 43r, 43u, 44d]*
82. **Proposition.** A design passing a design room's direction poll becomes the incumbent, a paperless rival cannot block it, and the chair may delay it one meeting for a paper to arrive. A rival with a paper can be polled alongside it, the room can direct changes, and by default a TS merge proceeds with it unless an alternative is actively progressing. *[proved by 82a, 82b, 82c, 82d] [from 42a, 42k, 42l, 43q, 43d, 42n, 31, 30]*
    a. **Lemma.** A design that passes a direction poll in a design room is the incumbent on its topic until it fails a later direction poll there. *[from 31, 30]*
    b. **Lemma.** A rival without a paper does not block an active proposal making progress, though the design room chair may delay the proposal one meeting for the rival's paper. *[from 42a, 42k]*

c. **Lemma.** A rival with a paper can be polled alongside the incumbent, which SD-4 encourages, and the room can direct changes, which the author should make or defend declining. *[from 43d, 42l, 43q]*

d. **Lemma.** When a TS is ready to merge into the IS and no alternative is actively progressing, SD-4's default is to move forward with the proposal it has. *[from 42n]*

83. **Proposition.** A design can become the incumbent at its first direction poll, when it has had less review than at any later meeting. From then on, every rival and every later change to it bears a reversal cost that grows as more is built on it. *[proved by 83a, 83b] [from 9, 8, 31, 30, 82, 81, 33]*

    a. **Lemma.** When a proposal is first presented to the committee in a design room, the chair may take a direction poll on it at that meeting. *[from 62c, 43c, 30, 24]*

    b. **Lemma.** Adopting a rival in an incumbent's place, or later changing it, replaces all or part of what a design room approved in a direction poll, so each is a reversal. *[from 31, 30, 33]*

84. **Proposition.** A rival wins a voter from an incumbent only with an advantage worth more to that voter than their share of the reversal's cost. That hurdle grows with each decision built on the incumbent, which faces no such hurdle itself. *[proved by 84a, 84b] [from 4, 81, 82, 31, 33, 29]*

    a. **Lemma.** Keeping an incumbent removes or replaces nothing, so of the two choices only adopting a rival in its place is a reversal. *[from 83b, 33]*

    b. **Lemma.** A participant does better with a rival than with the incumbent only when the rival's advantage to them outweighs their share of the reversal's cost, which keeping the incumbent avoids. *[from 84a, 10, 4, 29]*

85. **Lemma.** A direction poll asks voters only to agree that work should continue, less than a forwarding poll asks, so it can pass where a forwarding poll would fail. Yet passing it in a design room makes the design the incumbent. *[proved by 85a, 85b] [from 30, 31, 43r, 4, 83]*

    a. **Lemma.** A participant who would vote Favor on sending a design to the next room prefers that work on it continue rather than stop. One who doubts the design as it stands can prefer that too, since the room can direct changes. *[from 30, 43r, 43q, 4]*

    b. **Lemma.** Passing a direction poll in a design room makes a design the incumbent, while passing one in a study group does not. *[from 31, 24]*

86. **Proposition.** If a poll places a proposal among the work the committee aims to include in the next IS, the proposal joins a category SD-4 has chairs take up first. A meeting's hours are fixed, so the time it then gets comes out of the time left for papers outside those categories. *[from 43v, 14]*

87. **Proposition.** Close study takes longer the bigger a design is, and a bigger design has more places where a defect can go unnoticed. For a reviewer whose stake does not grow with size, close study therefore stops being worth its time sooner on a big design than on a small one. *[proved by 87a] [from 7, 6, 8, 4, 28, 29]*

    a. **Lemma.** Studying a proposal affects its poll only through what the participant learns, which can change their vote or what they say in discussion. So what studying brings them through that poll is worth no more than their stake in it. *[from 6, 43n, 28]*

88. **Proposition.** Participants who do not study a proposal typically abstain, so one costing more to study than most stakes in it is voted on mainly by those with large stakes. Unless those of them who object cast at least a third of the Favor-plus-Against votes, it can advance with no change at all. *[proved by 88a, 88b] [from 43g, 87, 7, 4, 63, 73]*
    a. **Lemma.** A participant whose stake in a proposal is worth less than the time studying it takes does better not studying it. *[from 87a, 7, 4, 29]*
    b. **Lemma.** When fewer than a third of the Favor-plus-Against votes are Against, Favor votes are more than twice the Against votes, which meets the 2:1 norm. *[from 43f, 26]*
89. **Proposition.** A design normally needs the 2:1 norm to advance, and once adopted, a paper or ballot comment to remove it normally needs the same, switching the default to keeping it. After a successful final ballot, design changes are expected to have near-unanimous consent, ISO bars technical changes at approval, and SD-4 calls published features much harder to update incompatibly. *[proved by 89a, 89b] [from 43f, 42a, 47e, 47j, 54c, 45g, 33]*
    a. **Lemma.** Removing an adopted design is a design change, so a proposal to do it needs a paper, and a national body can also request it in a ballot comment. *[from 42a, 47e, 62e]*
    b. **Lemma.** Design rooms own selecting designs and plenary does not change them, so a removal, by paper or comment, normally needs the 2:1 norm in a design room. *[from 43m, 44e, 43f, 89a]*
90. **Proposition.** An objector who arrives after a design has been adopted faces three obstacles at once. Removal needs a paper meeting the 2:1 norm, reversal costs more for every decision built on the design, and the record keeps its case but unwritten objections only in memory. *[proved by 90a, 90b, 90c] [from 89, 81, 84, 76, 78]*
    a. **Lemma.** Once a design is adopted, removing it normally needs a paper or ballot comment meeting the 2:1 norm, which switches the default to keeping it. *[from 89]*
    b. **Lemma.** Removing or replacing an adopted design is a reversal, whose cost grows with every decision built on it. A rival wins a voter only with an advantage worth more than their share of that cost. *[from 81, 84]*
    c. **Lemma.** The design's case lasts in its paper, while objections no one wrote down survive only with those who heard them. *[from 76]*

In plain terms: a design can become the incumbent at its first direction poll, when it has had less review than at any later meeting, and every rival or later change then bears a reversal cost that grows with each decision built on it. A proposal that costs more to study than most stakes in it is voted on mainly by those with large stakes, and it can advance with no change at all unless its objectors cast at least a third of the Favor-plus-Against votes. After adoption, removing a design normally needs the same 2:1 norm it needed to advance, and an objector who arrives then also faces the grown reversal cost and a record that keeps the design's case but unwritten objections only in memory.

6. Sponsors, proposal size, and parts

91. **Proposition.** A sponsor adds a voter to a room for each participant it pays to attend, and neither SD-4 nor the searched ISO clauses limit how many it pays. SC22 may cap only the experts each national body or liaison appoints, and at SC22 itself each national body counts once. *[from 60, 58e, 43b, 2, 52e, 52i, 52h, 53a]*
92. **Proposition.** Changing the features a sponsor's interests depend on costs that sponsor rework of what it built on them, lost fit with its requirements, and more review, all still to come. The gain of a better design is spread across all the feature's users, and the same 2:1 norm decides every proposal, however many users will rely on it. *[from 4, 10, 7, 17, 43f, 22]*
93. **Induction.** Each part of a design adopted in parts becomes work every later part builds on, so a rival to the whole design must replace every part already adopted. *[proved by 93a, 93b] [from 62a, 81, 33]*
    a. **Lemma.** An author can propose a design in papers that each propose part of it, and each later part builds on the parts already adopted. *[from 62a, 10]*
    b. **Lemma.** Adopting a rival to the whole design in its place replaces every part already adopted, a reversal of each, which redoes or discards the work built on it. *[from 33, 81]*
94. **Proposition.** Presenting a design in parts lets the first parts be adopted before the whole is reviewed, and each adopted part adds to what a rival to the whole must replace. The cost is a paper and a presentation for each part. *[proved by 94a, 94b, 94c] [from 62a, 43r, 43u, 44d, 93, 84, 7, 42h]*
    a. **Lemma.** Each part's paper advances on its own consensus calls in the design and wording rooms and at plenary, so an early part can be adopted before later parts are presented. *[from 62a, 43r, 43u, 44d]*
    b. **Lemma.** Each adopted part is one more part a rival to the whole must replace, raising the advantage the rival needs to win a voter. *[proved by 94b.i, 94b.ii] [from 93, 84]*
        i. A rival to the whole design must replace every part already adopted, so each adopted part adds work its reversal must redo or discard. *[from 93, 33]*
        ii. A rival wins a voter only with an advantage worth more than that voter's share of the reversal's cost, so a costlier reversal raises the advantage it needs. *[from 84, 94b.i]*
    c. **Lemma.** Each part needs its own paper, which takes time to write, and its own presenter at each meeting where it is considered. *[from 7, 42h]*
95. **Proposition.** Reviewers cannot study what later parts need of a first part that fixes a design's structure, SD-4 asks no paper to say, and once adopted, every rival must replace it. So a design-bound author's best first paper is the smallest part that fixes that structure and still meets the 2:1 norm against opponents who foresee the rest. *[proved by 95a, 95b, 95c, 95d] [from 9, 58l, 93, 63]*
    a. **Lemma.** A design-bound author does better when its first part fixes the features its gain depends on, as the structure later parts build on, than when a later part fixes them. The first part is adopted sooner, and every later part built on it adds to the cost of the reversal a rival would need. *[proved by 95a.i, 95a.ii, 95a.iii] [from 38, 94, 93, 81, 84, 4, 29]*

i. Parts advance one paper at a time, so the first part can be adopted before any later part, fixing its features sooner. *[from 94]*

ii. Every later part builds on the first, so each adds to the cost of the reversal that changing the first part's features would need. *[from 93, 81, 84]*

iii. A design-bound author's gain depends on those features staying fixed, so it does better when they are fixed sooner and changing them needs a costlier reversal. *[from 95a.i, 95a.ii, 38, 4, 29]*

b. **Lemma.** At the first part's polls, only those who foresee the rest can base their vote on what the later parts will need of it. Others cannot study that in detail until those parts are presented, and SD-4 asks no paper to state it. *[from 9, 6, 58l]*

c. **Lemma.** Against votes on the first part can keep it below the 2:1 line, while removing it once adopted normally needs a paper or ballot comment meeting the norm. So an opponent who foresees the rest does better voting Against the first part than waiting to vote Against a later part. *[proved by 95c.i, 95c.ii, 95c.iii] [from 63, 89, 4, 29]*

i. Against votes making up at least a third of the Favor-plus-Against votes on the first part keep it from meeting the 2:1 norm. *[from 63]*

ii. Once the first part is adopted, removing it normally needs a paper or ballot comment that itself meets the 2:1 norm. *[from 89]*

iii. Blocking the first part takes a third of the Favor-plus-Against votes, while removing it later takes over two thirds, so the opponent does better voting Against early. *[from 95c.i, 95c.ii, 63, 4, 29]*

d. **Lemma.** Of two first parts that fix the same features and meet the 2:1 norm against opponents who foresee the rest, the smaller takes less time to write and to review. So a design-bound author whose gain depends on those features does better with the smaller, which can fix them sooner and at less cost. *[from 7, 14, 38, 4, 29]*

96. **Proposition.** A large proposal has more sponsored authors in the room than a small one, and more meetings where a poll can make it the incumbent or end that standing. Each step it advances brings it nearer to work that can make plenary progress, which SD-4 has chairs take up first. *[from 36, 35, 31, 91, 43v, 43r, 43u, 44d]*

97. **Proposition.** A small proposal's author bears every meeting's cost alone, and outside SD-4's priority categories the proposal gets agenda time only after the priority work. Where WG21 runs study groups that incubate early proposals, such as EWGI and LEWGI, SD-4's statement that study groups own incubating applies to them. *[from 35, 1, 43v, 42d, 43k, 24]*

98. **Proposition.** Each result about large or small proposals rests on an attribute their definitions name: who pays the authors' costs, the number of authors, review's expected span, or the paper's length. So it holds for any proposal with that attribute, whatever its size. *[from 35, 36, 96, 97]*

In plain terms: a sponsor adds a voter to a room for each participant it pays to attend, and neither SD-4 nor the searched ISO clauses limit how many it pays. A design-bound author's best first paper is the smallest part that fixes the design's structure and still meets the 2:1 norm against opponents who foresee the rest. A large proposal has more sponsored authors in the room than a small one, whose author bears every

meeting's cost alone, and each such result holds for any proposal with the attribute it rests on.

7. Study groups and deadlocks

99. **Proposition.** On a topic with a study group, rival designs are first polled there, among those who chose that room over the rooms meeting with it. The design it sends on is where the design room starts, and the design room, which owns selecting designs, can still poll a rival that has a paper. *[proved by 99a, 99b, 99c] [from 43m, 43d]*
    a. **Lemma.** SD-4 has the convenor form a study group on a design room chair's recommendation when a topic has broad interest, no unified initial proposal, and a strong chair candidate. It gives the study group the role of producing an initial proposal that can then enter the design room. *[from 43l, 43k]*
    b. **Lemma.** A participant can attend only one of the rooms meeting at the same time, and votes and speaks only there. So a study group's voters are those for whom its topic outweighs the rooms meeting with it. *[from 16, 4, 28, 43b]*
    c. **Lemma.** The design room starts from the study group's initial proposal, which a rival without a paper does not block while it is an active proposal making progress. *[from 99a, 99b, 42k, 43k]*
100. **Proposition.** Paying participants to attend a study group puts a sponsor's voters where the design the design room starts from is chosen. Where the other voters cast fewer Favor-plus-Against votes, fewer paid participants make up a third of the total, enough to block, or over two thirds, enough to meet the norm. *[proved by 100a, 100b, 100c] [from 99, 91, 63]*
    a. **Lemma.** A group casts at least a third of the Favor-plus-Against votes exactly when it casts at least half as many as all the other voters together. *[from 26]*
    b. **Lemma.** A group casts more than two thirds of the Favor-plus-Against votes exactly when it casts more than twice as many as all the other voters together. *[from 26]*
    c. **Lemma.** A proposal meets the 2:1 norm whenever more than two thirds of its Favor-plus-Against votes are Favor, since its Favor votes then number more than twice its Against votes. *[from 43f, 26]*
101. **Proposition.** When each of two groups casts at least a third of the Favor-plus-Against votes against the other's design, neither design meets the 2:1 norm. The deadlock lasts until voters change position, a design changes, the voters present change, a chair calls consensus below the norm, or the design room sends both to competing TSs. *[proved by 101a, 101b] [from 63, 43f, 43r, 43m, 54i]*
    a. **Lemma.** The count in a poll depends only on who is present, what the poll asks, and how each person present votes on it, if at all. *[from 43b, 26]*
    b. **Lemma.** A design leaves a design room only when the room's chair calls consensus on the design and its ship vehicle. *[from 43r, 43m, 44d]*
102. **Proposition.** In a long deadlock, a group whose sponsors pay its costs bears less per meeting than a group paying its own way. In the case the table below describes, the first holds and the second yields, whichever design is better. *[proved by 102a, 102b, 102c, 102d, 102e, 102f, 102g] [from 2, 1, 101]*

a. **Lemma.** When one deadlocked group yields, the other holds, and no one else votes Against the holder's design, the deadlock ends with that design advancing. The holder gains its whole stake, the yielder none of its own, and neither bears the cost of holding through a deadlock. *[proved by 102a.i, 102a.ii, 102a.iii] [from 101, 63, 28]*
    i. A group that yields stops voting Against the other's design, so when no one else votes Against it, the holders' Favor votes meet the 2:1 norm. *[from 101, 63]*
    ii. The design that advances brings the holder the difference it makes to what the holder values, its whole stake, and brings the yielder none of what its own design would. *[from 28, 102a.i]*
    iii. The deadlock then ends at the next poll, so neither group bears the cost of holding through further meetings. *[from 102a.i, 1]*
b. **Lemma.** When both groups hold and the deadlock ends with either design equally likely, each can expect half its stake on average, less its cost of holding through it. Members whose costs sponsors pay bear less of that cost per meeting. *[from 101, 1, 2, 28]*
c. **Lemma.** Each group does better holding while the other yields, gaining its whole stake at no cost of holding, than when both yield to a compromise worth half its stake. *[proved by 102c.i, 102c.ii] [from 102a, 4, 29]*
    i. Holding while the other yields brings a group its whole stake at no cost of holding, while both yielding brings a compromise worth half its stake. *[from 102a]*
    ii. A whole stake is worth more than half of it, so each group ranks holding while the other yields above both yielding. *[from 102c.i, 4, 29]*
d. **Lemma.** Each group does better when both yield to a compromise worth half its stake than when both hold. That holds because a deadlock both hold brings half its stake on average, less a cost of holding. *[proved by 102d.i, 102d.ii] [from 102b, 1, 4, 29]*
    i. When both hold, a group can expect half its stake on average, less its cost of holding through the deadlock, a cost above zero since attending costs money and time. *[from 102b, 1]*
    ii. Half its stake with no cost of holding is worth more than half its stake less such a cost, so each group ranks both yielding above both holding. *[from 102d.i, 4, 29]*
e. **Lemma.** Each group does better when both yield to a compromise worth half its stake than when it yields while the other holds, which brings none of its stake. *[proved by 102e.i, 102e.ii] [from 102a, 4, 29]*
    i. Yielding while the other holds brings a group none of its stake, and both yielding brings it a compromise worth half its stake, with no cost of holding either way. *[from 102a]*
    ii. Half its stake is worth more than none, so each group ranks both yielding above yielding while the other holds. *[from 102e.i, 4, 29]*
f. **Lemma.** A group whose cost of holding is less than half its stake does better holding through a deadlock both hold than yielding while the other holds. That holds because half its stake on average, less that cost, leaves it better off than none of its stake. *[proved by 102f.i, 102f.ii] [from 102b, 102a, 4, 29]*

   i. Holding through a deadlock both hold brings half its stake on average less its cost of holding, and yielding while the other holds brings none of its stake. *[from 102b, 102a]*
   ii. With that cost less than half its stake, the first is worth more than nothing, so the group ranks holding above yielding. *[from 102f.i, 4, 29]*

g. **Lemma.** A group whose cost of holding is more than half its stake does better yielding while the other holds than holding through a deadlock both hold. That holds because half its stake on average, less that cost, leaves it worse off than none of its stake. *[proved by 102g.i] [from 102b, 102a, 4, 29]*
   i. With that cost more than half its stake, half its stake on average less that cost is worth less than nothing, so the group ranks yielding above holding. *[from 102f.i, 4, 29]*

Table 3. Two groups in a deadlock, with equal stakes in their own designs. Group A's members have their costs paid by sponsors and group B's pay their own, so holding through the deadlock costs group A less than half its stake and group B more than half. A deadlock both hold ends with either design equally likely, and both yielding means a compromise worth half its stake to each. First number: group A's rank. Second number: group B's rank. 4 is best. † marks a Pareto-optimal cell.

	B holds	**B yields**
A holds	2, 1	**4, 2 †**
A yields	1, 4 †	3, 3 †

[from 102c, 102d, 102e, 102f, 102g]

In plain terms: on a topic with a study group, the design room starts from the design the study group sends on, and where the other voters there cast fewer Favor-plus-Against votes, fewer paid participants are enough to block or to meet the norm. When each of two groups casts at least a third of the Favor-plus-Against votes against the other's design, neither design meets the 2:1 norm, and the deadlock lasts until positions, designs, or the voters present change, a chair calls consensus below the norm, or the design room sends both to competing TSs. In a long deadlock, a group whose sponsors pay its costs bears less per meeting than a group paying its own way, and in the case the table describes, the first holds and the second yields, whichever design is better.

8. Chairs, the convenor, and plenary

103. **Proposition.** Every consensus call has a cost: declaring consensus over Against votes risks later national body opposition, and declining delays work SD-4 says subgroup polls should favor. Departing from the 2:1 norm can cost the caller's post: chairs serve at the convenor's choice with no fixed term, and the convenor for terms SC22 may decline to renew. *[proved by 103a, 103b] [from 43e, 44c, 54h, 49b, 62e, 43a, 52d, 4, 25]*

a. **Lemma.** SD-4 asks a participant with a serious objection to a subgroup decision to raise it within their national body, which can oppose that decision at plenary or on a ballot. *[from 49b, 62e, 44c, 54h, 25]*

b. **Lemma.** Design and wording rooms forward a proposal, and plenary adopts one, only on a consensus call, so declining the call keeps the proposal at its current stage. *[from 43r, 43u, 44d, 24]*

104. **Proposition.** Above the 2:1 line after discussion, declaring consensus follows SD-4's norm, though a Neutral majority can be concerning, and at or below it SD-4 allows a re-poll after discussion. A chair whose stake is worth less than what departing can cost makes the call the norm gives. *[proved by 104a, 104b] [from 43f, 63, 43h, 103]*

    a. **Lemma.** A design room chair whose stake is worth less than what departing can cost ranks declaring consensus above declining or re-polling on a count above 2:1 after discussion. This holds when Neutral votes are not a majority of those voting. *[proved by 104a.i, 104a.ii, 104a.iii] [from 43f, 103, 4, 29]*

        i. On a count above 2:1 after discussion, the 2:1 norm is met, and when Neutral votes are not a majority of those voting, declining or re-polling departs from it. *[from 43f, 26]*

        ii. Declining or re-polling on such a count can cost the chair their post. *[from 104a.i, 103]*

        iii. What declining or re-polling can gain the chair over declaring comes from the poll's outcome, so it is at most their stake in the poll. *[from 28]*

    b. **Lemma.** On a count at or below 2:1 after discussion, such a chair ranks declining or re-polling above declaring consensus. *[proved by 104b.i, 104b.ii, 104b.iii] [from 43f, 103, 4, 29]*

        i. On a count at or below 2:1 after discussion, the 2:1 norm is unmet, so declaring consensus departs from it, while declining or a re-poll, which SD-4 allows, does not. *[from 43f, 26]*

        ii. Declaring consensus on such a count can cost the chair their post. *[from 104b.i, 103]*

        iii. What declaring consensus can gain the chair over declining or re-polling comes from the poll's outcome, so it is at most their stake in the poll. *[from 28]*

Table 4. A design room chair's consensus call, when the chair's stake in the proposal is worth less to them than what departing from SD-4's norm can cost them, and Neutral votes are not a majority of those voting. Columns: the count after discussion of the Against voters' concerns. Each cell: the chair's rank of the call. 2 is best within each column.

	Above 2:1	**Not above 2:1**
Declare consensus	**2**	1
Decline or re-poll	1	**2**

[from 104a, 104b]

105. **Proposition.** Of two papers otherwise alike, a chair does better taking up first the one whose author has a record of following procedures and responding to review, which promises fewer problems. That record comes only from taking part. *[proved by 105a, 105b] [from 3, 4, 14, 43v, 29]*

a. **Lemma.** A meeting's hours are fixed, and SD-4 has chairs reach some papers only as time allows, so a paper taken up first can get time a later one does not. *[from 14, 43v]*
b. **Lemma.** SD-4 expects an author to respond to review by taking part in the design room, presenting the proposal, answering questions, and making the changes the room directs. *[from 43q]*

106. **Proposition.** A participant who thinks a new proposal should not take up extended subgroup time can tell the chair privately and in advance. The chair may then time-box it and poll whether to go on, and nothing in SD-4 requires a record of the request. *[from 43o, 43p, 58g]*
107. **Proposition.** At plenary, design stays with the design rooms, objections not escalated in time get no weight, and all present are asked for serious design concerns. Most polls pass by unanimous consent, and when opposition is significant the convenor usually asks whether Against votes are personal or national body positions. *[proved by 107a, 107b, 107c, 107d] [from 44b, 44e, 78, 44a, 44c, 59]*
    a. **Lemma.** Plenary decides whether to adopt the changes the design and wording rooms recommend, and does not change a design or hold extended design discussion. *[from 59, 44d, 44e]*
    b. **Lemma.** Objections not escalated by 5pm the evening before plenary, on the email lists and within a national body, get no weight there. *[from 78]*
    c. **Lemma.** Design room chairs present the designs approved at the meeting, asking all present for any serious design concerns the subgroups may not have found. *[from 44a]*
    d. **Lemma.** Plenary first asks whether anyone objects to unanimous consent, which most polls pass, and on significant opposition the convenor usually asks whether Against votes are personal or national body positions. *[from 44b, 44c]*

In plain terms: every consensus call has a cost, and departing from the 2:1 norm can cost the caller's post, so a chair whose stake is worth less than what departing can cost makes the call the norm gives. Of two papers otherwise alike, a chair does better taking up first the one whose author has a record of following procedures and responding to review, a record that comes only from taking part. At plenary, design stays with the design rooms, objections not escalated in time get no weight, and most polls pass by unanimous consent.

9. Users, quality, and disclosure

108. **Proposition.** No user votes, and taking part costs the participant or their sponsor, while a better design's gain is spread across all users and seldom turns on one user's effort. Users' interests reach decisions only through what participants themselves value, papers a participant presents, TS feedback, and national bodies, which form positions only where that is worth their members' time. *[proved by 108a, 108b] [from 60, 62g, 17, 4, 51e]*
    a. **Lemma.** Apart from guests, whose choices the proof takes as given, participants cast the votes in WG21's polls and national bodies cast those in the ballots on its drafts. *[from 60, 44b, 59, 61, 18, 52i]*

b. **Lemma.** A national body forms a position through its members' discussion, as when its chair caucuses with them at a meeting, so each position takes time they could give to papers. *[from 49b, 14]*

109. **Proposition.** Fees are barred and everyone in a subgroup's room has one vote, yet only those who bear the cost of taking part, or whose sponsors do, vote in WG21's polls. ISO can remove inactive national body experts, and participants propose for needs they and their sponsors know, which, since users differ from them in what matters, need not match users'. *[from 55b, 51c, 51b, 47a, 52i, 52k, 60, 1, 2, 4, 17, 42h]*
110. **Proposition.** An author gains at adoption, while a flaw that implementation and use have not yet exposed costs users only after adoption, and can fall on any user of the feature. *[from 4, 8, 17, 21, 23]*
111. **Lemma.** Implementation and use reduce uncertainty about how well a design works, though even then a room cannot fully judge it at a poll. SD-4 describes a TS or white paper as a way to gain that experience, noting that its users typically should not rely on it in production code. *[from 8, 45a, 45b]*
112. **Proposition.** Winning a consensus call and producing the design that serves users best coincide only as far as voters can judge that at the poll and weigh it above other values. SD-4 assigns no one to judge it for them. *[proved by 112a, 112b] [from 4, 8, 59, 72, 111, 58f]*
    a. **Lemma.** A vote reflects which design serves users best only as far as the voter can judge that at the poll and weighs a good result for users above other values. *[from 4, 8, 111]*
    b. **Lemma.** Chairs call consensus on the votes of the participants present, counted under the 2:1 norm, and the consensus test asks nothing about whether an objection is right. *[proved by 112b.i, 112b.ii] [from 59, 72]*
        i. The participants present decide a design's content, as chairs call consensus on their votes counted under the 2:1 norm. *[from 59]*
        ii. Under SD-4's definition, none of the three conditions under which an objection prevents consensus asks whether the objection is right. *[from 72]*
113. **Proposition.** Close study costs the reviewer, while the gain of a better design is spread across all its users. So review worth more to users together than it costs, but less to any one participant, is review no participant does better doing. *[from 7, 17, 4, 87, 28, 29]*
114. **Proposition.** Papers approved separately at one meeting first come together when the project editor merges their wording, which the editor's report records and a pre-ballot editing committee verifies. No SD-4 step checks whether the designs work together, while each author gains from their own design's adoption and a conflict costs users and later rework. *[proved by 114a, 114b, 114c] [from 46b, 46a, 46c, 58k, 4, 8, 17, 10]*
    a. **Lemma.** Plenary approves each wording paper in its own poll, and during the meeting each paper's wording is written against the drafts adopted at its start. *[from 46b, 46a]*
    b. **Lemma.** After the meeting the project editor applies all approved changes and reports on merging them, and before a ballot an editing committee verifies they were applied correctly. *[from 46c]*

c. **Lemma.** Each author gains from their own design's adoption. A conflict between designs may surface only once they are implemented and used, and then it costs users and forces rework of what was built on them. *[from 4, 8, 17, 10]*

115. **Proposition.** Stating a drawback in the paper before an early poll can cost Favor votes when incumbency is decided, while a better-informed vote's gain is spread across the design's users. SD-4 says a good paper should show the alternatives considered, but it asks no paper to state its design's drawbacks, and a paper is considered without them. *[proved by 115a] [from 6, 83, 17, 4, 42c, 58i, 42f, 42h]*
    a. **Lemma.** SD-4 relies on the presenter, an expert in the paper's domain, to educate a room of general C++ experts. It also counts a feature's designers among the best experts on any concern about it. *[from 42i, 48a]*
116. **Proposition.** No rule requires a written answer to an objection, apart from an author's defense of declining a directed change and ISO's response to each enquiry ballot comment. For the author, a written answer makes the objection easier to raise again, but also shows later what would not be new, the test SD-4 applies to repeated objections. *[proved by 116a, 116b, 116c, 116d] [from 70, 58h, 43q, 54g, 54f, 13, 77, 47h]*
    a. **Lemma.** SD-4 asks no proposal paper to answer an objection, apart from the defense and new information an author who declines a directed change should bring. Its request that a good paper show the alternatives considered names no objection that must be among them. *[from 58h, 43q, 42c]*
    b. **Lemma.** ISO requires responses to all comments on an enquiry ballot, with written observations on each, while it only notes comments on a final draft for future consideration. *[from 54g, 54f, 54e]*
    c. **Lemma.** A written answer to an objection states the objection it answers, and a clear written objection is easier to raise again. *[from 13]*
    d. **Lemma.** A written answer shows later readers what a repeated objection must add to bring new information. It stays available after those present have left, and records what the objection and its answer contained. *[from 13, 47h]*

In plain terms: no user votes, and users' interests reach decisions only through what participants value, papers a participant presents, TS feedback, and national bodies, while a better design's gain is spread across all users. Winning a consensus call and producing the design that serves users best coincide only as far as voters can judge that at the poll and weigh it above other values, and SD-4 assigns no one to judge it for them. SD-4 asks no paper to state its design's drawbacks, and no SD-4 step checks whether designs approved in separate papers work together.

10. The schedule and national bodies

117. **Proposition.** A proposal that misses an edition's feature cutoff waits at least for the next edition, planned three years later. It can ship sooner as a white paper, which has no prescribed schedule, and the same schedule keeps an unfinished feature from delaying the features that are ready. *[proved by 117a] [from 45f, 44d, 15, 32]*
    a. **Lemma.** Once an edition's draft goes to ballot, SD-4 treats a ballot comment that requests adding a feature as out of scope, and WG21 plans the next edition three years later. *[from 47f, 32, 15]*
118. **Proposition.** At the feature cutoff, a one-meeting delay for a paperless rival sends a proposal to the next planned edition. So does an Against vote keeping it below the 2:1 line, when the chair follows the norm. *[proved by 118a, 118b] [from 117, 42k, 63, 43f, 44d, 32]*
    a. **Lemma.** Plenary adopts only changes the design and wording rooms recommend, so a proposal is approved for an edition only if those rooms forward it by the edition's feature cutoff. *[from 44d, 43r, 43u, 32]*
    b. **Lemma.** When a chair follows the 2:1 norm, their room forwards a proposal only on a count above 2:1 after discussion. *[from 43f, 26, 104, 104b, 43r, 43u, 24]*
119. **Proposition.** A TS delays IS adoption by about one to two years, and its experience can bring incompatible changes first: a gain to result-seeking authors, a risk to design-bound ones. The design room can still require a TS, and ISO and SC22 now recommend the lighter white paper. *[proved by 119a, 119b] [from 42m, 45a, 45b, 45g, 43m, 45e, 45f, 111, 37, 38, 4]*
    a. **Lemma.** An incompatible change that fixes a shortcoming found in implementing and using a TS makes the adopted design serve its users better, which raises a result-seeking author's gain. *[from 45a, 45b, 111, 37, 4]*
    b. **Lemma.** An incompatible change made after implementing and using a TS can alter the features a design-bound author's gain depends on keeping fixed. *[from 45a, 45b, 38]*
120. **Lemma.** Every national body may vote on enquiry and final ballots, which pass when two thirds of the P-members' votes are positive and at most a quarter of all votes negative, not counting abstentions or negative votes without technical comments. ISO encourages a body without national consensus or expert input to abstain, sustained opposition does not apply to these votes, and a final draft that fails goes back to SC22. *[from 52a, 54a, 54h, 54b, 51f, 53e, 54d, 25]*
121. **Proposition.** A national body's No opposes every feature in the document at once, including ones its own experts support, while WG21 considers its comments whatever its vote. A body valuing the rest of a document above blocking one feature does better voting Yes with comments, and one wanting the work in no form does better voting No. *[proved by 121a, 121b, 121c, 121d] [from 59, 52f, 47c, 54g, 54e, 120, 4, 29]*
    a. **Lemma.** A body that values the rest of a document above blocking any one feature ranks Yes with comments above abstaining. That holds because only the Yes counts toward the approval it prefers, and its comments are considered either way. *[proved by 121a.i, 121a.ii] [from 120, 47c, 4]*
        i. Abstentions are not counted, so of the two only Yes with comments adds to the positive P-member votes approval needs. *[from 120]*

   ii. WG21 considers a body's comments whatever its vote, so a body preferring approval gains nothing by abstaining and ranks Yes with comments above it. *[from 47c, 121a.i, 4]*

b. **Lemma.** Such a body ranks abstaining above No, since only the No counts against the approval it prefers and its comments gain nothing from a No. *[proved by 121b.i, 121b.ii] [from 120, 47c, 4]*
   i. A No with technical comments counts against approval, while an abstention is not counted at all. *[from 120]*
   ii. Its comments are considered whatever its vote, so for a body preferring approval a No only lowers approval's chance, and it ranks abstaining above No. *[from 47c, 121b.i, 4]*

c. **Lemma.** A body that wants the work in no form ranks No above abstaining, since only the No counts against approval, which is what SD-4 says No is for. *[proved by 121c.i, 121c.ii] [from 120, 47b, 4]*
   i. SD-4 says a No vote should express that the national body does not want the work to progress at all in any form. *[from 47b]*
   ii. Of No and abstaining, only the No counts against approval, so a body that wants the work in no form ranks No first. *[from 121b.i, 121c.i, 4]*

d. **Lemma.** Such a body ranks abstaining above Yes with comments, since only the Yes counts toward approval of work it wants in no form. *[proved by 121d.i] [from 120, 47b, 4]*
   i. Of abstaining and Yes with comments, only the Yes counts toward approval, so a body that wants the work in no form ranks abstaining first. *[from 121a.i, 4]*

Table 5. A national body's ballot vote. Columns: how the body values the document; "Values the rest" means it values the rest of the document above blocking any one feature in it, and "Opposes the work" means it does not want the work at all. Each cell: its rank of the vote. 3 is best within each column.

	Values the rest	Opposes the work
Yes with comments	**3**	1
Abstain	2	2
No	1	**3**

[from 121a, 121b, 121c, 121d]

122. **Proposition.** At ballot, only a national body that values blocking one feature above the whole rest of the document does better voting No over it. SD-4 expects such a body to have told the committee all along, and ISO asks every national body to make its standpoint clear early. *[from 121, 47b, 47i, 41e, 51e, 4, 29]*

123. **Proposition.** A national body forms its position by its panel's own rules, which SD-4 does not state, so whether a sponsor's paid participants add weight there depends on those rules. In the United States, INCITS gives a voting member's representatives one vote, apart from up to four Defense Department votes, and bars consultants a voting member funds from voting membership. *[proved by 123a] [from 49d, 49b, 2]*
    a. **Lemma.** INCITS, whose Executive Board approves the United States position that goes to ANSI, gives each organization one voting membership, apart from up to four Department of Defense votes on technical committees, admits a consultant as a voting member only if no voting member funds it, and lets the representatives of standards developing organizations and consortia be employed by other voting members. A United States delegation may also decide a new issue at an international plenary by caucus consensus. *[from 57a, 57b, 57c, 57d, 57e]*

In plain terms: at the feature cutoff, a one-meeting delay for a paperless rival, or an Against vote keeping a proposal below the 2:1 line when the chair follows the norm, sends it to the next edition, planned three years later. A national body's No opposes every feature in a document at once, so only a body that values blocking one feature above the whole rest of the document does better voting No over it. Whether a sponsor's paid participants add weight to a national body's position depends on its panel's own rules.

11. The settled pattern

124. **Lemma.** Whatever the others choose, a regular does no better than to vote Favor on colleagues' proposals they support, and on one they doubt, Against does best only when worth more than its added cost. Otherwise Neutral, not voting, or even Favor can do best, since Favor risks nothing with the colleague, while Against costs risk to colleagues' support and time for reasons, and changes the result only one or two votes above the line. *[proved by 124a, 124b, 124c] [from 64, 65, 63, 66, 67, 68, 77, 118, 5, 4, 62, 29, 34]*
    a. **Lemma.** A Favor vote on a colleague's proposal gives that colleague no reason to support the voter's own proposals less readily. SD-4 names no step in which a chair asks a Favor voter for reasons. *[from 5, 27, 43h, 58b]*
    b. **Lemma.** On a colleague's proposal that a regular supports, Favor does at least as well for the regular as any other vote, whatever the others choose. *[proved by 124b.i, 124b.ii] [from 4]*
        i. Favor brings a proposal one vote nearer to meeting the 2:1 norm, Neutral or no vote leaves the comparison unchanged, and Against raises by two the Favor votes it needs. *[from 43f, 26, 63a, 63b]*
        ii. Favor costs the regular no standing and no time for reasons, so no other vote costs the regular less. *[from 124a]*
    c. **Lemma.** When an Against vote on a doubted proposal is worth less to the regular than its added cost, the regular's best vote is Neutral, not voting, or Favor. Favor is best when whatever standing a Neutral vote or none risks is worth more to the regular than keeping one Favor vote from the proposal. *[proved by 124c.i, 124c.ii] [from 65a, 4, 29]*

    i. Neutral or no vote leaves the 2:1 comparison unchanged and draws no request for reasons. Neither gives the colleague more reason than an Against vote to support the regular less readily. *[from 63b, 58b, 5]*
    ii. Favor adds one to the Favor count but costs the regular no standing and no time for reasons. *[from 26, 124a]*

125. **Lemma.** Whatever the others choose, a design-bound author does no better than to seek direction polls as early as they can pass, propose first the smallest part that fixes the features its gain depends on, and seek the IS over a TS unless the room requires one. It does no better than to leave out trade-offs no reviewer would raise, ask for no poll against a rival it expects to lose, and answer objections with the cheapest change that keeps the 2:1 norm met, leaving those features alone unless the room directs otherwise. *[proved by 125a, 125b, 125c, 125d] [from 83, 81, 84, 85, 95, 94, 119, 115, 73, 69, 74, 92, 43q, 31, 38, 62a, 34]*
    a. **Lemma.** A design-bound author does no better than to seek direction polls as early as they can pass, since every change to an incumbent bears a reversal cost that grows. Passing one sooner makes the features its gain depends on costlier to change at every later meeting. *[from 83, 81, 84, 85, 31, 38]*
    b. **Lemma.** A design-bound author does no better than to leave out a trade-off no reviewer would raise. *[proved by 125b.i, 125b.ii]*
        i. Leaving it out costs the author nothing the rules impose, since SD-4 asks no paper to state its design's drawbacks and a paper is considered without them. *[from 115, 58i]*
        ii. Stating it takes time to write, can cost Favor votes before an early poll, and can lead the room to change the features the author's gain depends on. *[from 7, 115, 6, 43q, 38]*
    c. **Lemma.** A design-bound author does no better than to ask for no poll against a rival it expects to lose. *[proved by 125c.i, 125c.ii, 125c.iii]*
        i. A poll the author's design loses to a rival can make the rival the incumbent, or leave the author's design no longer the incumbent. *[from 31, 30]*
        ii. Either result lowers the chance that the features the author's gain depends on are adopted, and an incumbent rival puts a reversal cost on any return to them. *[from 84, 83, 38]*
        iii. The chair may take any polls they choose, so a request from the author can add a poll the chair would not otherwise take but cannot remove one. *[from 43c, 62a]*
    d. **Lemma.** A design-bound author does no better than to answer objections with the cheapest change that keeps the 2:1 norm met. That change leaves the features its gain depends on alone unless the room directs a change to them. *[proved by 125d.i, 125d.ii] [from 73, 69, 38]*
        i. Concerns stated as changes can be granted one at a time, so the author can grant those that leave the features its gain depends on unchanged. *[from 74, 38]*
        ii. Declining a change the room directs harms the proposal's chances unless the author defends the decision with new information. *[from 43q]*

126. **Lemma.** Whatever the others choose, a result-seeking author does no better than to state a trade-off and grant a change whenever the better design they can bring is worth more to it than the votes or

work they cost. It does no better than to seek an early poll or an early first part only where sooner adoption outweighs the higher cost this puts on later improvements, and to back a rival that would serve users better by more than its share of the reversal's cost. *[proved by 126a, 126b, 126c] [from 83, 81, 84, 94, 115, 73, 74, 37, 62a, 34]*

a. **Lemma.** A stated trade-off or a granted change brings a result-seeking author whatever it adds to how well the adopted design serves users. Its cost to that author is the Favor votes it loses and the work it takes. *[proved by 126a.i, 126a.ii] [from 37]*
    i. What participants learn from a paper can change their votes, so a stated trade-off can lead the room to direct a change, while a granted change alters the design directly. *[from 6, 43q]*
    ii. Either can cost Favor votes, stating a trade-off takes time to write, and a change takes the author work, more when it revises the design throughout. *[from 115, 6, 7, 11]*
b. **Lemma.** An early poll or an early first part can bring a result-seeking author adoption sooner. It also makes every later improvement bear a reversal cost that grows, and those improvements are part of that author's gain. *[proved by 126b.i, 126b.ii, 126b.iii] [from 83, 93, 81, 4, 37]*
    i. A direction poll taken early lets the later polls a design needs come sooner, and a first part can be adopted before the rest is presented. *[from 30, 43r, 94]*
    ii. Every later change to an incumbent, or to a design built on adopted parts, bears a reversal cost that grows as more is built on it. *[from 83, 93, 81]*
    iii. A result-seeking author gains from how well the adopted design serves users, so later improvements, which that cost makes harder, are part of its gain. *[from 37, 4]*
c. **Lemma.** A rival that would serve users better brings a result-seeking author that much more gain if adopted, and backing it costs the author its share of the reversal's cost. *[proved by 126c.i, 126c.ii] [from 37]*
    i. As a participant, the author can back a rival by voting Favor on it or by asking the chair for a poll on it. *[from 21, 43b, 62a]*
    ii. Replacing an incumbent redoes or discards the work built on it, and the author bears its share of that cost. *[from 10, 33]*

127. **Lemma.** Between two options, an author who values both a good result for users and particular features chooses as the author type would whose value differs more between them. *[proved by 127a, 127b] [from 4, 37, 38, 29]*
    a. **Lemma.** When one option is better for users and in the features, or the options differ in only one of these, an author who values both takes the better option. The type whose value differs more makes the same choice. *[from 4, 37, 38]*
    b. **Lemma.** When one option is better for users and the other in the features, an author who values both weighs the two differences on its one scale. It takes the option better in the value that differs more, which is the choice of the type whose gain depends on that value. *[from 4, 37, 38, 29]*

128. **Proposition.** The rules treat both kinds of author alike: the same moves, and no request to state a design's drawbacks, a sponsor, or what the author's gain depends on. So stating a trade-off can cost a result-seeking author Favor votes that a design-bound author who leaves it out keeps. *[proved by 128a, 128b] [from 62a, 58i, 58j, 42f, 42h, 115, 125, 126]*
    a. **Lemma.** In the case the table below describes, a result-seeking author ranks stating the trade-off above leaving it out. That holds because stating it brings a design that serves users better while the proposal still meets the 2:1 norm. *[proved by 128a.i, 128a.ii] [from 37, 115, 6, 4, 29]*
        i. In that case, stating the trade-off leads the room to change the design so that it serves users better, which is what a result-seeking author's gain depends on. *[from 6, 43q, 37]*
        ii. In that case the proposal meets the 2:1 norm either way, so stating the trade-off costs no adoption, and with other values held equal the author ranks it first. *[from 128a.i, 4, 29]*
    b. **Lemma.** In that case, a design-bound author ranks leaving the trade-off out above stating it, since stating it brings a change to a feature its gain depends on. *[proved by 128b.i, 128b.ii] [from 38, 115, 6, 4, 29]*
        i. In that case, stating the trade-off leads the room to change a feature the design-bound author's gain depends on keeping fixed, which lowers that gain. *[from 6, 43q, 38]*
        ii. In that case the proposal meets the 2:1 norm either way, so leaving the trade-off out costs no adoption, and with other values held equal the author ranks it first. *[from 128b.i, 4, 29]*

Table 6. An author's choice, before an early direction poll, whether to state in the paper a trade-off no reviewer would otherwise raise, when stating it would lead the room to change a feature a design-bound author's gain depends on, in a way that serves users better, and the proposal would meet the 2:1 norm either way. Columns: the author's type. Each cell: the author's rank of the choice. 2 is best within each column.

	Result-seeking	Design-bound
State it	**2**	1
Leave it out	1	**2**

[from 128a, 128b]

129. **Lemma.** Whatever the others choose, a small proposal's author does no better than to keep it presented, in person, remotely where offered, or by another qualified and prepared presenter, while it is worth more to them than costs they bear alone. When those costs, which grow with each meeting it waits behind priority work, outweigh it, the author does no better than to let it stop. *[proved by 129a, 129b, 129c] [from 97, 80, 1, 51c, 42h, 42f, 62, 34]*
    a. **Lemma.** A small proposal is considered at a meeting only when it has an on-time paper and a presenter. The presenter can be its author, in person or remotely where that is offered, or another qualified and prepared presenter. *[from 42f, 42h, 51c, 35]*

b. **Lemma.** Each meeting at which the proposal waits behind priority work adds another meeting's cost to what its author bears alone. *[proved by 129b.i, 129b.ii] [from 97, 1, 129a]*
    i. Outside SD-4's priority categories, a small proposal gets agenda time only after the priority work, so it can wait through meetings without being reached. *[from 97]*
    ii. At each meeting it waits, it is considered only with an on-time paper and a presenter, and its author bears that meeting's cost alone. *[from 129a, 97, 1]*
c. **Lemma.** Keeping the proposal presented gives its author the proposal's chance of adoption at those costs, while letting it stop gives up that chance and ends them. *[from 129a, 129b, 4]*

130. **Lemma.** Whatever the others choose, a study group chair does no better than to poll the competing designs, side by side where there are several, and send on a design that meets the 2:1 norm, or none when none does. The chair does better departing from this only when their stake in the contest is worth more to them than what departing can cost them. *[proved by 130a, 130b] [from 99, 101, 102, 103, 43l, 43k, 43d, 43f, 62, 34]*
    a. **Lemma.** A study group chair's choice of which design leaves the group is a consensus call under the 2:1 norm. A chair whose stake is worth less than what departing can cost them makes the call that norm gives: a design that meets it, or none when none does. *[proved by 130a.i, 130a.ii] [from 62c, 24, 43k, 43f, 104]*
        i. A study group chair polls which design leaves the group as its initial proposal, and the 2:1 norm covers subgroup polls, so study group polls too. *[from 62c, 43k, 26, 24]*
        ii. A chair whose stake is worth less than what departing can cost them makes the call the norm gives, which here sends on a design that meets it, or none. *[from 104, 130a.i]*
    b. **Lemma.** When several designs compete, a study group chair whose stake is worth less than what departing can cost them does no better than to poll them side by side. *[proved by 130b.i, 130b.ii, 130b.iii]*
        i. SD-4 encourages polling competing designs side by side, so that the order of polls cannot lose the one with the most consensus. *[from 43d]*
        ii. Whatever order a chair sets instead, the change it can make to which design leaves the group is worth no more to the chair than their stake in the contest. *[from 28, 43c]*
        iii. Setting an order departs from what SD-4 encourages, and since the convenor appoints chairs with no fixed term, departing can cost a chair their post. *[from 43d, 43a, 103]*

131. **Lemma.** Whatever the others choose, a design room chair does no better than to declare consensus above the 2:1 line after discussion, when Neutral votes are not a majority, and to re-poll or decline at or below it, unless their stake is worth more than what departing can cost them. It does no better than to take up first, of papers otherwise alike, the one whose author has a record of following the procedures. *[proved by 131a, 131b] [from 104, 103, 105, 106, 85, 118, 122, 62, 34]*
    a. **Lemma.** For every count after discussion, a design room chair whose stake is worth less than what departing can cost them ranks first the call the 2:1 norm gives. With Neutral votes not a majority, that is consensus above the line and a re-poll or declining at or below it. *[from 104, 104a, 104b]*

b. **Lemma.** Of two papers otherwise alike, a design room chair does better taking up first the one whose author has a record of following the procedures, since it promises fewer problems. *[from 105, 3]*

132. **Lemma.** Whatever the others choose, the convenor does no better than to follow SD-4's plenary procedure unless their stake is worth more than what departing can cost them. That procedure has the convenor declare consensus when no one objects to unanimous consent, or when all opposition rests on objections not escalated in time and holds no national position. *[proved by 132a, 132b] [from 107, 78, 103, 44b, 49c, 44d, 41a, 62, 34]*

    a. **Lemma.** When all opposition rests on objections not escalated in time and holds no national position, SD-4's plenary procedure has the convenor declare consensus. *[proved by 132a.i, 132a.ii] [from 44b, 41a]*

        i. SD-4 says objections not escalated by the deadline should not be given weight in plenary. *[from 78, 49c]*

        ii. With that opposition given no weight and no national position behind it, every position the convenor gives weight to is Favor or Neutral. *[from 132a.i, 44d, 44c]*

    b. **Lemma.** The change that departing from SD-4's plenary procedure can make to an outcome is worth to the convenor no more than their stake in it. Departing can also cost the convenor their post, since SC22 appoints the convenor for terms of up to three years and may appoint another candidate when one ends. *[from 28, 103, 52d]*

133. **Lemma.** Whatever the others choose, a national body does no better than to vote Yes with comments if it values a document's rest above blocking any one feature, and No otherwise. It does no better than to raise its objections early, as SD-4 and ISO ask, and ISO encourages it to abstain when it lacks national consensus or expert input. *[proved by 133a, 133b] [from 121, 122, 120, 123, 77, 41e, 51e, 62, 34]*

    a. **Lemma.** A national body for which blocking some one feature is worth at least as much as the rest of the document does no better than to vote No. *[proved by 133a.i, 133a.ii]*

        i. A national body ballots on the whole document, so such a body does no worse when the document fails than when it passes with that feature. *[from 59, 4]*

        ii. WG21 considers its comments whatever its vote, and only a No with technical comments counts against approval, so No does at least as well for it as abstaining or Yes. *[from 47c, 120, 133a.i]*

    b. **Lemma.** A national body does no better than to raise its objections early. *[proved by 133b.i, 133b.ii, 133b.iii]*

        i. An objection raised early comes while less has been built on the design, so a change it wins costs less than the same change later. *[from 81]*

        ii. An objection first raised at plenary without escalation gets no weight there. *[from 78]*

        iii. At the final ballot ISO accepts no technical changes, so an objection first raised there can change the outcome only by failing the whole document. *[from 54c, 120]*

134. **Lemma.** Whatever the others choose, a sponsor does no better than to pay for participants in the rooms whose decisions it values, a study group included. That holds as long as the votes they add are worth their cost to it. *[proved by 134a] [from 91, 100, 96, 123, 1, 2, 62, 34]*
    a. **Lemma.** A sponsor cannot vote in a participant's place, so the votes it can add to a room are those of participants it pays, each at a cost to it. Each participant it pays values the sponsor's interests among the other things they value. *[from 2, 1, 4, 91, 22]*
135. **Lemma.** Whatever the others choose, a user does no better than to take the move, if any, with the largest positive net gain: taking part, TS feedback, a paper a participant presents, or a request to a national body. Taking part costs one person the full price while a better design's gain is shared by all users, so it is that move only for a user whose own stake outweighs that cost. *[proved by 135a, 135b] [from 108, 62g, 1, 17, 4, 29, 34]*
    a. **Lemma.** A user casts no vote. Each move open to a user costs the user time or money: taking part, TS feedback, a paper a participant presents, or a request to a national body. *[from 60, 62g, 1, 7]*
    b. **Lemma.** Taking part brings a user a net gain only when their own stake outweighs its full price, which they bear alone. *[proved by 135b.i, 135b.ii]*
        i. Sponsors pay the costs of participants, and a user is not one, so a user deciding to take part bears the full price of attending alone. *[from 22, 23, 1]*
        ii. Taking part can bring the user no more than the difference the outcome makes to what they value, which is their stake. A better design's gain is spread across all users of its feature. *[from 28, 4, 17]*
136. **Proposition.** The rules check these effects: every view must be heard, a third of the votes can block, rivals with papers can be polled together, and implementation experience informs votes. Plenary invites serious design concerns, national bodies vote on each balloted draft and have enquiry comments answered, editing committees verify applied changes, minutes can be challenged, and appeals are open. *[proved by 136a, 136b, 136c, 136d, 136e]*
    a. **Lemma.** SD-4 requires every view to be heard and understood and the majority to listen, and asks for respect, good faith, and no collusion, with conduct concerns reported through ISO's process. It welcomes negative comments, and any group casting a third of the Favor-plus-Against votes as Against keeps a proposal from meeting the 2:1 norm. *[from 40f, 41b, 43n, 40d, 40b, 40c, 40a, 63]*
    b. **Lemma.** Chairs may allow limited exploration of alternatives, and SD-4 encourages polling alternatives side by side and says a good paper shows the alternatives considered and its design principles. Followup and rebuttal papers can be considered, and an author who declines a directed change should defend that decision. *[from 42l, 43d, 42c, 42b, 42g, 43q]*
    c. **Lemma.** Competing TSs are possible, a TS or white paper should list the controversial issues to address before it merges, and SD-4 expects pre-merge issues to be addressed. Implementation experience informs voters, a TS can merge early for integration testing, a white paper has no prescribed schedule, and the schedule keeps an unfinished feature from delaying ready ones. *[proved by 136c.i, 136c.ii, 136c.iii, 136c.iv] [from 54i, 45h, 45c, 45d, 111, 45f, 117]*

i. ISO allows competing TSs that do not conflict with existing International Standards. SD-4 asks a TS or white paper to list its must-address controversial issues, and expects pre-merge issues to be addressed. *[from 54i, 45h, 45c]*

ii. Implementation and use reduce uncertainty about how well a design works, and SD-4 describes a TS or white paper as a way to gain that experience. *[from 111]*

iii. SD-4 encourages participants to keep pre-merge issues few, so that a TS or white paper can merge early for integration testing in the full context of the IS. *[from 45d]*

iv. A white paper has no prescribed schedule, and WG21's three-year schedule keeps an unfinished feature from delaying the features that are ready. *[from 45f, 117]*

d. **Lemma.** Every expert is appointed by a national body or liaison organization, SC22 may cap how many each appoints, and ISO recommends that experts keep their national body informed. ISO asks national bodies to take account of all interests concerned nationally, SD-4 asks design rooms to weigh direction group opinions on any proposal, and agenda topics need on-time papers. *[from 52e, 52h, 52g, 51e, 43j, 42e]*

e. **Lemma.** SD-4 asks those with a concern to tell the chairs and authors early, plenary solicits serious design concerns, every national body may vote on each enquiry and final ballot, every enquiry comment gets a written response, and an editing committee verifies that approved changes were applied correctly. Minutes can be challenged, appeals are open, on drafts only on questions of principle or harm to the reputation of ISO or IEC, ISO says consensus requires resolving substantial objections and ends the committee stage when all technical issues are resolved, and SD-4 works in addition to the Directives. *[from 48b, 44a, 120, 54f, 54g, 46c, 55c, 56a, 56b, 56c, 56d, 53i, 51d, 53j, 39a]*

137. **Theorem.** Under SD-4 and the quoted clauses of ISO/IEC Directives Part 1, in the edition replacing the JTC 1 Supplement, and given the axioms, these choices form a settled pattern. In it, no player improves their position by changing only their own choice. A design-bound author seeks direction polls as early as they can pass, proposes first the smallest part fixing the features it needs, and seeks the IS over a TS. It leaves out trade-offs no reviewer would raise and answers objections with the cheapest change that keeps the 2:1 norm met. A result-seeking author states trade-offs and makes changes whenever the better design they bring is worth their cost to it. It backs a rival that serves users better by more than its share of the reversal's cost. An author who values both chooses, at each two-way choice, as the type whose value differs more. A small proposal's author keeps it presented while it is worth costs borne alone. A regular votes Favor on colleagues' proposals they support, and Against one they doubt only when that is worth more than its added cost. Otherwise the regular takes whichever of Neutral, not voting, or Favor does best, and Favor risks nothing with the colleague. A sponsor pays for participants where the decisions it values are made, study groups included. Chairs and the convenor follow SD-4's procedure unless their stake outweighs what departing can cost them. A national body votes Yes with comments unless blocking one feature is worth as much to it as the rest of the document. A user takes part only when their own stake outweighs a cost borne alone, and otherwise uses a cheaper move or none.

In this pattern a design's content can be fixed when it is least reviewed. Its first favorable poll can make it the incumbent, and every later change or rival then pays a reversal cost that grows whether or not the design is wrong. Objections count only in the forms and times that attending teaches. In a long deadlock, the group whose sponsors pay its costs can hold longest, whichever design is better. A proposal that costs more to study than most participants' stakes in it can advance with no change at all. An objector who arrives after adoption meets these obstacles together. Other things equal, a proposal advances more readily when its authors keep presenting it or know the procedures, or when a sponsor pays for participants who hold its position. The same holds when the chair expects fewer problems from it, a poll has placed it among the work aimed at the next IS, or it is already the incumbent. No user votes, and the needs that reach a vote are those participants and their sponsors know, which need not match those of users at large. The participants present decide content, and SD-4 sets no limit on how many voters one sponsor can pay for. A change to a design costs the sponsors who have built on it, while its gain is spread across all users. An author gains at adoption, while a hidden flaw costs users later. Technical merit enters through what participants value, including a good result for users, and through the discussion, papers, and implementation experience that inform their votes. SD-4 assigns no one to rule on it, and the consensus test never asks whether an objection is right. No SD-4 step checks whether separately adopted designs work together, review whose gain is spread across all users goes undone, and the record need not show whether objections were answered. The rules ask no author to state a trade-off. Leaving one out costs a design-bound author nothing the rules impose, while stating one can cost a result-seeking author votes. A first part that fixes a structure is reviewed before anyone can study what the later parts need of it, and every rival must then replace it. The rules also contain checks on each of these effects, and each result about large or small proposals holds for any proposal with the attributes it rests on. *[proved by 137a, 137b, 137c, 137d, 137e, 137f, 137g, 137h, 137i, 137j, 137k, 137l] [from 5, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 59, 82, 83, 81, 84, 79, 88, 90, 86, 105, 102, 108, 109, 91, 92, 72, 110, 112, 111, 114, 113, 70, 115, 95, 98, 61, 51a, 58f, 42h, 12, 4, 34]*

a. **Lemma.** Given the others' choices, each choice the Theorem names does at least as well for its player as any other move the rules give it. So no player improves their position by changing only their own choice, and the choices form a settled pattern. *[proved by 137a.i, 137a.ii, 137a.iii] [from 61, 62]*
    i. Whatever the others choose, each author and each regular does no better than the choices the Theorem names for them, a small proposal's author included. *[from 125, 126, 127, 129, 124]*
    ii. Whatever the others choose, study group chairs, design room chairs, the convenor, national bodies, sponsors, and users do no better than the choices the Theorem names for them. *[from 130, 131, 132, 133, 134, 135]*
    iii. A choice doing at least as well as any other, given the others' choices, leaves its player nothing to gain by changing only that choice. *[from 34, 29]*

b. **Lemma.** In this pattern a design's content can be fixed when it is least reviewed. Every later change or rival then pays a reversal cost that grows whether or not the design is wrong. *[proved by 137b.i, 137b.ii, 137b.iii]*
    i. A design-bound author in this pattern seeks direction polls as early as they can pass, and a direction poll can pass where a forwarding poll would fail. *[from 125, 85]*
    ii. A design can become the incumbent at its first direction poll in a design room, when it has had less review than at any later meeting. *[from 82, 83]*
    iii. From then on, every rival and every later change bears a reversal cost that grows as more is built on the design, whether or not it is wrong. *[from 83, 81]*
c. **Lemma.** Objections count only in the forms and times that attending teaches. An objector who arrives after adoption meets these obstacles together: removal needs a paper meeting the 2:1 norm, reversal costs grow, and unwritten objections survive only in memory. *[proved by 137c.i, 137c.ii]*
    i. Objections need fixed forms and times, such as escalation before plenary's deadline, and participants learn these rules partly by attending, some only that way. *[from 79, 78]*
    ii. Removing an adopted design needs a paper meeting the 2:1 norm, reversal costs more for every decision built on it, and unwritten objections survive only in memory. *[from 90]*
d. **Lemma.** In a long deadlock, the group whose sponsors pay its costs can hold longest, whichever design is better. A proposal that costs more to study than most participants' stakes in it can advance with no change at all. *[proved by 137d.i, 137d.ii]*
    i. A group whose sponsors pay its costs bears less per meeting of a long deadlock than a group paying its own way. In the case the deadlock table describes, the first holds and the second yields, whichever design is better. *[from 102, Table 3]*
    ii. A proposal costing more to study than most stakes in it is voted on mainly by those with large stakes. Unless those of them who object cast at least a third of the Favor-plus-Against votes, it can advance with no change at all. *[from 88]*
e. **Lemma.** Other things equal, a proposal advances more readily when its authors keep presenting it or know the procedures, or when a sponsor pays for participants who hold its position. The same holds when the chair expects fewer problems from it, a poll has placed it among the work aimed at the next IS, or it is already the incumbent. *[proved by 137e.i, 137e.ii, 137e.iii, 137e.iv, 137e.v, 137e.vi]*
    i. A proposal is considered only while an author or another qualified and prepared presenter presents it, so an author who keeps presenting it keeps it able to advance. *[from 42h, 62a]*
    ii. Other things equal, an author who knows the procedures has the better chance of adoption, and that knowledge comes partly from attending. *[from 79, 12]*
    iii. Each participant a sponsor pays to attend adds a voter, and when those voters hold a proposal's position, their Favor votes bring it nearer the 2:1 norm. *[from 91, 43f, 26]*
    iv. Of papers otherwise alike, a design room chair in this pattern takes up first the one they expect fewer problems from, whose author has a record of following the procedures. Since a

meeting's hours are fixed, a paper taken up first can get time a later one does not. *[from 131, 105, 3, 14, 43v]*

v. A poll placing a proposal among the work aimed at the next IS puts it in a category SD-4 has chairs take up first. *[from 86]*

vi. An incumbent faces no reversal hurdle, while a rival wins a voter from it only with an advantage worth more than that voter's share of the reversal's cost. *[from 84]*

f. **Lemma.** No user votes, and the needs that reach a vote are those participants and their sponsors know, which need not match those of users at large. The participants present decide content, and SD-4 sets no limit on how many voters one sponsor can pay for. *[proved by 137f.i, 137f.ii, 137f.iii, 137f.iv]*

i. No user votes, and users' interests reach decisions only through what participants value, papers a participant presents, TS feedback, and national bodies. *[from 60, 108]*

ii. Participants propose for needs they and their sponsors know, which need not match users' needs, since users differ from them in what matters. *[from 109]*

iii. The participants present decide a design's content: chairs call consensus on their votes, counted under the 2:1 norm. *[from 59]*

iv. A sponsor adds a voter for each participant it pays to attend, and SD-4 sets no limit on how many it pays. Each must still be an expert a national body or liaison appoints, and SC22 may cap how many each appoints. *[from 91, 52e]*

g. **Lemma.** A change to a design costs the sponsors who have built on it, while its gain is spread across all users. An author gains at adoption, while a hidden flaw costs users later. *[proved by 137g.i, 137g.ii]*

i. Changing the features a sponsor's interests depend on costs that sponsor rework of what it built on them, while a better design's gain is spread across all the feature's users. *[from 92]*

ii. An author gains at adoption, while a flaw that implementation and use have not yet exposed costs users only after adoption. *[from 110]*

h. **Lemma.** Technical merit enters through what participants value, including a good result for users, and through the discussion, papers, and implementation experience that inform their votes. SD-4 assigns no one to rule on it, and the consensus test never asks whether an objection is right. *[proved by 137h.i, 137h.ii, 137h.iii]*

i. Each participant values a good result for users, what they learn from a paper or its discussion can change their vote, and implementation and use reduce uncertainty about a design. *[from 4, 6, 111]*

ii. Winning a consensus call and producing the design that serves users best coincide only as far as voters can judge that at the poll and weigh it above other values. *[from 112]*

iii. SD-4 assigns no one to rule which side of a technical dispute is correct, and none of the consensus test's three conditions asks whether an objection is right. *[from 58f, 72]*

i. **Lemma.** No SD-4 step checks whether separately adopted designs work together, review whose gain is spread across all users goes undone, and the record need not show whether objections were answered. *[proved by 137i.i, 137i.ii, 137i.iii]*

   i. SD-4 names no step in which anyone checks whether designs adopted in separate papers work together. *[from 58k, 114]*

   ii. Review worth more to users together than it costs, but less to any one participant, is review no participant does better doing, so in this pattern no participant does it. *[from 113, 137a, 34]*

   iii. The record SD-4 and the searched ISO clauses require of a WG21 poll need not show whether an objection was answered or outvoted. *[from 70]*

j. **Lemma.** The rules ask no author to state a trade-off. Leaving one out costs a design-bound author nothing the rules impose, while stating one can cost a result-seeking author votes. *[proved by 137j.i, 137j.ii, 137j.iii]*

   i. The rules give both kinds of author the same moves and ask no author to state a design's drawbacks, and SD-4 never uses the word trade-off. *[from 128, 58i]*

   ii. Leaving a trade-off out costs a design-bound author nothing the rules impose, since SD-4 asks no paper to state its drawbacks and considers a paper without them. *[from 125b.i, 115]*

   iii. Stating a trade-off can cost a result-seeking author Favor votes that a design-bound author who leaves it out keeps. *[from 128, 115]*

k. **Lemma.** A first part that fixes a structure is reviewed before anyone can study what the later parts need of it, and every rival must then replace it. *[from 95, 9]*

l. **Lemma.** The rules also contain checks on each of these effects, and each result about large or small proposals holds for any proposal with the attributes it rests on. *[proved by 137l.i, 137l.ii]*

   i. The rules check these effects through hearing every view, the one-third block, polling rivals together, implementation experience, plenary's call for design concerns, ballots, editing committees, minutes, and appeals. *[from 136]*

   ii. Each result about large or small proposals rests on an attribute their definitions name, so it holds for any proposal with that attribute, whatever its size. *[from 98]*

In plain terms: when each player makes the choices the rules reward, given what they value, no player improves their position by changing only their own choice, and in the settled pattern those choices form, a design's content can be fixed when it is least reviewed. The rules ask no author to state a trade-off, so leaving one out costs a design-bound author nothing the rules impose, while stating one can cost a result-seeking author votes. The rules also contain checks on each of these effects.

12. Corollaries

138. **Corollary.** If the settled pattern holds, the record should show the following. SD-4 keeps meeting records password-protected but lets poll questions and numeric results be quoted, and each prediction names its evidence, counter-evidence, an innocent explanation, and the data telling them apart. *[proved by 138a, 138b, 138c, 138d, 138e, 138f, 138g, 138h, 138i, 138j, 138k] [from 50a]*

a. **Corollary.** Where one sponsor pays both a proposal's authors and many of the participants present, and its interests include the proposal's adoption, Against votes are a smaller share of the Favor-plus-Against votes than on other proposals; poll results and stated affiliations, taken as the observable measure of who pays, are the evidence, and an Against share no lower would count against it. An innocent explanation is that colleagues share technical views, and comparing their votes with their votes on proposals of similar design on the same topic from authors with other affiliations, whose adoption the sponsor's interests do not include, would tell the two apart. *[proved by 138a.i, 138a.ii, 138a.iii, 138a.iv, 138a.v, 138a.vi] [from 91, 4, 65, 63, 137]*
    i. A participant whose costs the sponsor pays values the sponsor's interests, so they weigh the proposal's adoption as a gain against any doubt they have about it. *[from 4, 22]*
    ii. Since an Against vote raises the Favor votes needed, such a participant needs a larger doubt than one without that sponsor before an Against vote is worth its added cost. *[from 138a.i, 63a, 65, 124]*
    iii. The sponsor adds one such voter for each participant it pays to attend, so the more of those present it pays, the smaller the Against share of the Favor-plus-Against votes. An Against share no lower than on other proposals would therefore count against the prediction. *[from 91, 138a.ii, 137]*
    iv. SD-4 lets poll questions and numeric results be quoted publicly, so poll results show the Against share. It asks no author to state who pays their costs, so who pays can be observed only where someone states it, as a stated affiliation does. *[from 50a, 58j]*
    v. Participants sharing the authors' technical views support their proposals, and in the settled pattern a regular votes Favor on proposals they support, so shared views also yield few Against votes. *[from 4, 124]*
    vi. Votes that follow shared views depend on the design, so they would be the same on a similar design on the same topic from authors with other affiliations. The sponsor's interests do not include that proposal's adoption, so the model predicts more Against votes from those participants there, and the comparison tells the two apart. *[from 138a.v, 138a.i, 138a.ii]*
b. **Corollary.** Among proposals with similar stakes, the share of those present not voting rises with paper length; poll results, attendance, and lengths are evidence, and no rise would count against it. An innocent explanation is that longer papers cover narrower topics fewer participants know, and comparing papers on topics of similar breadth but different length would tell the two apart. *[proved by 138b.i, 138b.ii, 138b.iii, 138b.iv, 138b.v] [from 87, 43g, 137]*
    i. A longer paper takes more time to study, so among proposals with similar stakes, close study of a longer one is worth its time to fewer participants. *[from 7, 6, 87, 28]*
    ii. Participants who did not study a paper typically do not vote on its polls, so the share of those present not voting rises with length. No rise would count against the prediction. *[from 43g, 138b.i, 137]*
    iii. Poll results give the votes cast, attendance gives the number present, and the paper gives its length, so together they give the share not voting at each length. *[from 50a, 13]*

iv. If longer papers cover narrower topics that fewer participants know, those unfamiliar typically do not vote, so narrowness alone would also make the share not voting rise with length. *[from 43g]*

v. Among papers on topics of similar breadth, narrowness predicts no rise with length while study time still grows with length, so that comparison tells the two apart. *[from 138b.iv, 138b.i, 138b.ii]*

c. **Corollary.** Large proposals appear as sequences of papers, each building on parts already adopted; the mailings are the evidence, and large designs adopted in one paper would count against it. An innocent explanation is that large designs are easier to review in parts, and whether parts are published together or each only after the last is adopted tells them apart. *[proved by 138c.i, 138c.ii, 138c.iii, 138c.iv, 138c.v, 138c.vi] [from 93, 94, 95, 96, 137]*

i. A large proposal's review is expected to span more than one edition, so presenting it in parts lets its first parts be adopted sooner than the whole could be. *[from 36, 94]*

ii. In the settled pattern a design-bound author proposes first the smallest part that fixes the features it needs, and each adopted part adds to what a rival must replace. *[from 125, 93, 94]*

iii. So in the settled pattern large proposals appear as sequences of papers, each building on parts already adopted, and large designs adopted in one paper would count against the prediction. *[from 138c.i, 138c.ii, 4, 96, 137]*

iv. Every significant proposal requires a paper and a published paper stays available, so the mailings show whether a large design arrived in one paper or in parts. *[from 42a, 13, 42d]*

v. If large designs are easier to review in parts, that ease would also yield sequences of papers, and reviewing in parts is as easy when the parts are published together. *[from 7]*

vi. For a design-bound author, part of the gain from parts is that each is adopted before reviewers can study what later parts require of it. Publishing the parts together forgoes that gain, so the model predicts each part published only after the last is adopted, and the timing tells the two apart. *[from 9, 95, 94, 138c.v]*

d. **Corollary.** The first papers of designs later extended part by part describe their scope as small and seldom set out the later parts they serve; the mailings are the evidence, and first papers that set out their later parts would count against it. An innocent explanation is that reviewers prefer small steps and a small first step does lower risk, and checking whether later parts relied on choices in the first that it did not describe would tell the two apart. *[proved by 138d.i, 138d.ii, 138d.iii, 138d.iv, 138d.v, 138d.vi, 138d.vii] [from 95, 115, 125, 137]*

i. In the settled pattern a design-bound author's first paper is the smallest part that fixes the structure its gain depends on, so it describes a small scope. *[from 95, 125]*

ii. SD-4 asks no paper proposing part of a design to state what later parts will require of it, so a first paper is considered without that statement. *[from 58l, 42f, 42h]*

iii. Setting out the later parts lets reviewers of the first weigh what it commits the design to, which can cost Favor votes at the poll that makes it the incumbent. *[from 6, 9, 83, 115]*

iv. So first papers of designs extended part by part seldom set out their later parts, and first papers setting them out would count against the prediction. *[from 138d.i, 138d.ii, 138d.iii, 125, 137]*

v. A published paper stays available, so the mailings show what each first paper described and what its later parts went on to propose. *[from 13, 42d]*

vi. A preference for small steps and the lower risk of a small step also yield small first papers, but give no reason to leave out choices later parts rely on. *[from 8]*

vii. The model has the first paper fix the structure later parts need without setting them out, so later parts relying on choices the first did not describe favor the model. Their absence favors the innocent explanation. *[from 138d.i, 138d.ii, 138d.vi, 95]*

e. **Corollary.** After a design passes a design room's direction poll, papers and poll questions on its topic present rivals as changes to it, even when no technical content has changed; poll questions and paper titles are the evidence, and rivals presented as equal alternatives would count against it. An innocent explanation is that the wording tracks the incumbent's real change of status; comparing like rivals in same-form polls before and after incumbency tells them apart. *[proved by 138e.i, 138e.ii, 138e.iii, 138e.iv, 138e.v] [from 82, 83, 84, 43d, 137]*

i. Incumbency comes from passing a direction poll, not from any change in content, and adopting a rival afterward replaces a design whose continuation the room approved, which is a reversal. *[from 31, 30, 33]*

ii. Every rival bears a reversal cost the incumbent does not, so a poll on a rival is a poll on changing the incumbent, even when the rival's content is unchanged. The model predicts papers and poll questions presenting rivals as changes, and rivals presented as equal alternatives would count against the prediction. *[from 138e.i, 82, 83, 84, 137]*

iii. SD-4 lets poll questions be quoted publicly and a published paper stays available, so poll questions and paper titles show how rivals are presented. *[from 50a, 13]*

iv. If the wording only tracks the incumbent's status, then in polls of the same form a like rival draws the same support before and after incumbency. *[from 43d]*

v. The model has each voter weigh their share of a reversal's cost only once a design is the incumbent, so it predicts less support for a like rival afterward. The comparison therefore tells the two apart. *[from 84, 81, 138e.iv]*

f. **Corollary.** Removals after adoption are rare, can come through a paper or a national body comment, and are more likely when national bodies whose No votes could fail a ballot oppose the design; working drafts, ballot comments, and poll results are the evidence, and frequent removals by narrow margins would count against it. An innocent explanation is that adopted designs rarely need removal, and comparing removals with the defect reports filed against adopted features would tell the two apart. *[proved by 138f.i, 138f.ii, 138f.iii, 138f.iv, 138f.v] [from 89, 81, 121, 120, 137]*

i. Removing an adopted design takes a paper or a national body's ballot comment, and normally the 2:1 norm, against a reversal cost that grows with each decision built on it. So removals are rare, and frequent removals by narrow margins would count against the prediction. *[from 89, 81, 84, 47d, 137]*

ii. When national bodies whose No votes could fail a ballot oppose a design, keeping it risks the whole document, a cost removal avoids, so removal is then more likely. *[from 120, 121, 122, 54d, 4]*

iii. Editors' reports summarize changes applied to working drafts, ballot comments state national body positions, and poll results are quotable, so together they show each removal, its route, and its margin. *[from 46c, 47d, 50a]*

iv. If adopted designs rarely need removal, removals would be rare even with no hurdle to removing them. *[from 89]*

v. Defect reports filed against adopted features show how often removal might be needed. The model predicts rare removals even where such reports are many, while the innocent explanation predicts few reports wherever removals are few. *[from 138f.i, 138f.iv, 8]*

g. **Corollary.** Small proposals stop advancing without a negative poll more often than large proposals do; the mailings and poll results are the evidence, and no such difference would count against it. An innocent explanation is that small proposals are more often dropped on reflection, and comparing self-funded and sponsored authors' stops on papers of similar length would tell the two apart. *[proved by 138g.i, 138g.ii, 138g.iii, 138g.iv, 138g.v] [from 129, 97, 96, 137]*

i. A small proposal's author bears every meeting's cost alone, and those costs grow with each meeting it waits behind priority work. Once they outweigh its worth to the author, the author does no better than to stop presenting it, so it stops advancing with no negative poll. *[from 129, 97, 42h, 42f]*

ii. The sponsors of a large proposal pay its authors' costs, and each step it advances brings it nearer the work chairs take up first. So small proposals stop advancing without a negative poll more often than large ones, and no such difference would count against the prediction. *[from 138g.i, 36, 96, 137]*

iii. The mailings show when a proposal's revisions stop, and quotable poll results show whether a negative poll came first. *[from 13, 42d, 50a]*

iv. If small proposals are more often dropped on reflection, that alone would also make them stop more often without a negative poll, whoever pays their authors. *[from 35]*

v. In the model, stops depend partly on who bears the costs, so among papers of similar length it predicts more stops by self-funded authors than by sponsored ones. Reflection alone predicts no such difference, so the comparison tells the two apart. *[from 98, 138g.i, 138g.ii, 138g.iv]*

h. **Corollary.** National bodies state positions on few proposals; national body comments and plenary records of national positions are the evidence, and positions on most proposals would count against it. An innocent explanation is that national bodies agree with most proposals, and

comparing how often they state positions on proposals with and without Against votes in subgroups tells them apart. *[proved by 138h.i, 138h.ii, 138h.iii, 138h.iv, 138h.v, 138h.vi] [from 78, 108, 49b, 14, 120, 137]*

   i. A national body forms a position only where that is worth its members' time, and by default an objection raised during a meeting is not a national position. *[from 108, 78, 49b]*
   ii. A ballot asks a national body to vote on a whole document, so it can vote without forming a position on each proposal in the document. *[from 120, 59]*
   iii. Time spent forming positions during a meeting comes out of its fixed hours, so in the settled pattern national bodies state positions on few proposals. Positions on most proposals would count against the prediction. *[from 14, 138h.i, 138h.ii, 133, 137]*
   iv. Ballot comments are national body positions, and when plenary opposition is significant the convenor usually asks whether Against votes are national positions. So comments and plenary records show where national bodies stated positions. *[from 47d, 44c]*
   v. If national bodies agree with most proposals, they would also state few positions, since a Yes vote expresses agreement without one. *[from 120]*
   vi. Agreement predicts positions more often on proposals drawing Against votes in subgroups, while the cost of forming them keeps positions rare there too, so the comparison tells the two apart. *[from 138h.v, 138h.i, 138h.iii]*

i. **Corollary.** Papers that advance over Against votes often leave unstated the objections they faced, beyond the alternatives and declined directions SD-4 asks papers to address; the papers and poll results are the evidence, and papers that routinely state and answer their objections would count against it. An innocent explanation is that the objections were answered in discussion, and comparing such papers with the answers subgroup chairs give at plenary when asked would tell the two apart. *[proved by 138i.i, 138i.ii, 138i.iii, 138i.iv, 138i.v] [from 70, 116, 58h, 7, 137]*

   i. A proposal can advance with objections unresolved, and a written answer costs the author time and makes the objection easier to raise again. *[from 69, 116, 7, 13]*
   ii. SD-4 asks proposal papers to address only the alternatives considered and declined directions, so in the settled pattern papers advancing over Against votes often leave their other objections unstated. Papers routinely stating and answering them would count against the prediction. *[from 58h, 138i.i, 125, 137]*
   iii. A published paper stays available and poll results are quotable, so together they show which papers advanced over Against votes and which objections each states. *[from 13, 50a]*
   iv. If the objections were answered in discussion, the answers would exist outside the papers, with those present and possibly in the protected meeting records. *[from 76, 50a]*
   v. Asked at plenary, a subgroup chair summarizes whether an item was considered and the reason for the subgroup's decision. Answers there to objections the papers omit favor the innocent explanation, and their absence favors the model. *[from 44f, 73, 138i.iv]*

j. **Corollary.** If some objections left unresolved point to real defects, features adopted while Against votes remained show more defects found after adoption than features adopted by unanimous

consent; issues lists, correction papers, removals, and poll results are the evidence, and no difference would count against it. An innocent explanation is that contested features are more novel or more used, so more people look for their defects, and comparing features matched on scope, novelty, implementation experience before adoption, and use would tell the two apart. *[proved by 138j.i, 138j.ii, 138j.iii, 138j.iv, 138j.v] [from 69, 73, 112, 8, 137]*

i. A proposal can advance with objections unresolved, and the cheapest response keeping the 2:1 norm met is no change or a partial change. So a feature adopted while Against votes remained can keep what its objectors pointed to. *[from 69, 73]*

ii. A feature adopted by unanimous consent drew no Against vote at that poll, and defects can go unnoticed until a design is implemented and used. So if some unresolved objections point to real defects, features adopted while Against votes remained show more defects found after adoption than those, and no difference would count against it. *[from 138j.i, 44b, 8, 112, 137]*

iii. Issues lists, correction papers, and removals record defects found after adoption, and quotable poll results show which features were adopted while Against votes remained. *[from 42a, 46b, 50a]*

iv. If contested features are more novel or more used, more people look for their defects, so more would be found in them even if no objection pointed to one. *[from 8, 6]*

v. Matching features on scope, novelty, implementation experience before adoption, and use holds those causes fixed, so a remaining difference favors the model and none favors the innocent explanation. *[from 138j.iv, 138j.ii, 111]*

k. **Corollary.** If closer study before adoption finds more defects, topics on which competing papers from separate author groups reached the deciding poll show fewer defects after adoption than topics with one paper; author lists, mailings, issues lists, and removals are the evidence, and no difference would count against it. An innocent explanation is that important or well-understood topics draw competing papers and expert attention anyway, and comparing topics of equal novelty and poll turnout tells them apart. *[proved by 138k.i, 138k.ii, 138k.iii, 138k.iv, 138k.v, 138k.vi] [from 87, 113, 4, 43d, 43g, 137]*

i. A competing paper's author values their own proposal's adoption, which defects found in the rival can help win. Close study of the rival is therefore worth more to them than to most participants. *[from 4, 6, 28, 87]*

ii. Close study worth less to any one participant than its cost goes undone, so a topic with one paper can lack study that a rival author's stake would supply. *[from 113, 138k.i]*

iii. So if closer study before adoption finds more defects, topics whose deciding poll had competing papers from separate author groups show fewer defects after adoption than one-paper topics. No difference would count against the prediction. *[from 138k.i, 138k.ii, 43d, 137]*

iv. Author lists and mailings show which topics had competing papers from separate author groups, and issues lists and removals record defects found after adoption. *[from 13, 42d, 42a]*

v. If important or well-understood topics draw competing papers and expert attention anyway, their importance or understanding alone would also leave fewer defects on topics with competing papers. *[from 6, 8]*

vi. Novelty measures how well understood a topic is, and turnout how much attention it drew, since those unfamiliar with a poll's material typically do not vote. So among topics equal in both, a remaining difference favors the model, and none favors the innocent explanation. *[from 43g, 138k.v, 138k.iii]*

139. **Corollary.** Each change below, to a rule or to what a move costs, would remove the support for a named item, and the model predicts the stated effect. *[proved by 139a, 139b, 139c, 139d, 139e, 139f, 139g, 139h, 139i]*

a. **Corollary.** If design polls were anonymous, the standing part of an Against vote's cost would shrink for voters who do not speak against the proposal. The chair could no longer ask Against voters for reasons, and the model predicts more Against votes on colleagues' proposals. *[proved by 139a.i, 139a.ii, 139a.iii] [from 64, 5, 65, 137]*

i. In an anonymous poll, a colleague cannot see how a voter who did not speak against the proposal voted, so that voter's Against vote risks less of the colleague's support. *[from 5, 64, 27]*

ii. Asking Against voters for reasons requires knowing who they are, so anonymity ends those requests, and with them the time an Against vote takes in preparing reasons. *[from 43h, 64, 7]*

iii. With both parts of an Against vote's added cost smaller, more doubts are worth that cost, so the model predicts more Against votes on colleagues' proposals. *[from 139a.i, 139a.ii, 65, 124, 137]*

b. **Corollary.** If a chair could ask voters of every position for reasons, Neutral would bear the reasons cost Against now bears, while not voting would bear none. The model predicts fewer Neutral votes and more participants not voting. *[proved by 139b.i, 139b.ii] [from 64, 63, 43g, 137]*

i. Under that rule a Neutral voter could be asked for reasons, which take time to prepare, while a participant who does not vote has no position to explain. *[from 64, 43g, 7]*

ii. Neutral and not voting leave the 2:1 comparison equally unchanged, so of the two only Neutral would bear the reasons cost. The model therefore predicts fewer Neutral votes and more participants not voting. *[from 63, 139b.i, 4, 137]*

c. **Corollary.** If the rules limited how many participants whose costs one sponsor pays may vote in a room, the votes a sponsor can add would be capped. The model predicts smaller differences in poll margins between large and small proposals. *[proved by 139c.i, 139c.ii, 139c.iii] [from 91, 96, 134, 137]*

i. A sponsor adds a voter for each participant it pays to attend, so limiting how many such participants may vote in a room caps the votes it can add. *[from 91]*

ii. A large proposal has more sponsored authors in the room than a small one, and each is a voter a sponsor added. So part of the difference in poll margins between large and small proposals comes from votes sponsors add. *[from 96, 91, 134]*

iii. Capping those votes shrinks that part, so the model predicts smaller differences in poll margins between large and small proposals. *[from 139c.i, 139c.ii, 137]*

d. **Corollary.** If the rules required a written answer to each objection in the record, the record would show whether an objection was answered. The model predicts that papers advancing over Against votes would state the objections they faced. *[proved by 139d.i, 139d.ii] [from 70, 116, 58h, 137]*
   i. Once the record must hold each objection and its answer, the objection is in writing whatever the paper says, and an answer already written takes little time to restate. *[from 7, 13]*
   ii. Leaving the objection out of the paper then saves the author little, while stating it there shows later readers what would not be new. So the model predicts that papers advancing over Against votes would state the objections they faced. *[from 139d.i, 116, 137]*

e. **Corollary.** If the rules set a decision procedure for deadlocks lasting a set number of meetings, a deadlock lasting that long would end by that procedure. The model predicts fewer deadlocks that end when one side stops attending. *[proved by 139e.i, 139e.ii, 139e.iii] [from 102, 101, 137]*
   i. Such a procedure ends a deadlock at the set number of meetings, so no group bears the cost of holding through it for more meetings than that. *[from 101, 1]*
   ii. A group whose cost of holding exceeds half its stake does better yielding, for example by no longer attending, than holding while the other group holds. That cost grows with each meeting a group holds, faster for a group paying its own way. *[from 102g, 102, 1]*
   iii. With the cost of holding capped, fewer groups' costs pass half their stakes before the procedure decides, so the model predicts fewer deadlocks that end when one side stops attending. *[from 139e.i, 139e.ii, 137]*

f. **Corollary.** If a TS did not delay adoption compared with an edition, the author's cost of accepting a TS would fall. The model predicts more designs sent to a TS or white paper. *[proved by 139f.i, 139f.ii] [from 119, 137]*
   i. If a TS did not delay adoption, the cost left to an author of accepting one would be the risk that experience brings incompatible changes first. *[from 119]*
   ii. That risk falls on design-bound authors, while the experience is a gain to result-seeking ones, so more authors would accept a TS or white paper. The model therefore predicts more designs sent there. *[from 139f.i, 119, 45e, 126, 125, 137]*

g. **Corollary.** If SD-4 adopted ISO's steps for assessing, registering, and addressing a sustained opposition, with a recorded answer, it would state a step for assessing and registering one. The model predicts registered oppositions in WG21's records. *[proved by 139g.i, 139g.ii, 139g.iii] [from 71, 70, 137]*
   i. Under ISO's steps, the leadership registers a declared opposition it judges not sustained, or sustained but not resolvable without compromising consensus, and continues the work. *[from 53f, 53h, 53g]*
   ii. A registered opposition is in writing, which makes it easier to raise again, at less cost to the objector than writing a paper. *[from 13, 7, 75]*

iii. Proposals can advance while objections remain unresolved, and declaring an opposition then gives an objector a written record, so the model predicts registered oppositions in WG21's records. *[from 69, 139g.i, 139g.ii, 70, 137]*

h. **Corollary.** If forming national positions during meetings cost less, national bodies would take more positions, and more significant plenary opposition would hold national positions, which the room's count does not settle. The model predicts larger changes in answer to objections national bodies back, slower progress for contested proposals, and less weight at plenary for how many participants each side has present. *[proved by 139h.i, 139h.ii, 139h.iii, 139h.iv] [from 78, 107, 132, 133, 73, 91, 123, 137]*

   i. A national body forms a position only where that is worth its members' time. So a lower cost of forming one during a meeting means positions on more proposals, and more plenary opposition holding them. *[from 108, 78, 107]*

   ii. SD-4's plenary procedure has the convenor declare consensus over opposition only when it holds no national position, so the room's count does not settle opposition that holds one. *[from 132, 107, 44c]*

   iii. Answering such opposition, rather than outvoting it, takes more than the cheapest change keeping the 2:1 norm met. So the model predicts larger changes in answer to objections national bodies back, and slower progress for contested proposals while those changes are made. *[from 139h.ii, 132, 73, 137]*

   iv. National positions form under each body's own rules, such as INCITS's one voting membership per organization, not by a count of the room. So with more of them, the model predicts less weight at plenary for how many participants each side has present. *[from 123, 123a, 91, 139h.ii, 137]*

i. **Corollary.** If forming national positions during meetings cost more, national bodies would take fewer, plenary would seldom meet opposition holding them, and the whole-document ballot would be their main check. Appeals would remain, on drafts only for questions of principle or harm to ISO's or IEC's reputation, and the model predicts attendance, paid participants, and early incumbency deciding more content. *[proved by 139i.i, 139i.ii, 139i.iii, 139i.iv] [from 59, 122, 132, 133, 91, 83, 62e, 56c, 137]*

   i. A national body forms a position only where that is worth its members' time, so a higher cost means fewer positions and less plenary opposition holding them. *[from 108, 78, 107]*

   ii. Besides positions at meetings, a national body's moves on content include comments, appeals, and its ballot vote on the whole document. Comments are considered whatever the vote, and work continues during an appeal, so the ballot vote is the move that counts directly toward stopping a document. *[from 62e, 121, 56d, 120]*

   iii. With fewer national positions, the convenor's procedure settles more plenary opposition by the room's count, in which a sponsor adds a voter for each participant it pays to attend. *[from 139i.i, 132, 107, 91]*

iv. A design first made the incumbent when least reviewed then meets fewer later checks, so the model predicts attendance, paid participants, and early incumbency deciding more content. *[from 83, 139i.ii, 139i.iii, 59, 137]*

In plain terms: if the settled pattern holds, the record should show a smaller Against share where one sponsor pays a proposal's authors and many participants present, large designs adopted in parts, rare removals, and small proposals that stop advancing without a negative poll, among other patterns. Each prediction names its evidence, what would count against it, and a comparison that tells it apart from an innocent explanation. Changes such as anonymous design polls, a limit on how many participants one sponsor pays may vote, or a required written answer to each objection would remove the support for named results, with effects the model predicts.